

Age and Learning Shapes Sound Representations in Auditory Cortex During Adolescence

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This **important** study suggests that adolescent mice exhibit less accuracy than adult mice in a sound discrimination task when the sound frequencies are very similar. The evidence supporting this observation is **solid** and suggests that it arises from cognitive control differences between adolescent and adult mice. The adolescent period is largely understudied, despite its contribution to shaping the adult brain, which makes this study interesting for a broad range of neuroscientists.

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Abstract

Adolescence is a developmental period characterized by heightened plasticity, yet how ongoing development affects sensory processing and cognitive function is unclear. We investigated how adolescent (postnatal day 20-40) and adult (postnatal day 60-80) mice differ in performance on a pure tone Go/No-Go auditory discrimination task of varying difficulty. Using dense electrophysiological recordings, we measured spiking activity at single neuron resolution in the auditory cortex while mice were engaged in the task. As compared to adults, adolescent mice showed lower auditory discrimination performance in the harder task. This performance difference was due to higher response variability and weaker cognitive control expressed as lower response bias. Adolescent and adult neuronal responses differed only slightly in representations of pure tones when measured outside the context of learning and the task. However, cortical representations after learning within the context of the task were markedly different. We found differences in stimulus- and choice-related activity at the single neuron level representations, as well as lower population-level decoding in adolescents. Overall, cortical decoding in adolescents was lower and slower, especially for difficult sound discrimination, reflecting immature cortical representations of sounds and choices. Notably, we found age-related differences, which were higher after learning, reflecting the combined impact of age and learning. Our findings highlight distinct neurophysiological and behavioral profiles in adolescence, underscoring the ongoing development of cognitive control mechanisms and cortical plasticity during this sensitive developmental period.

Introduction

Adolescence is a developmental stage characterized by the continued refinement of perception and cognition. During this period, the brain is highly sensitive to environmental influences, making it particularly vulnerable to both negative experiences, such as substance abuse, and positive influences, such as supportive relationships[1, 2]. While the adolescent brain exhibits greater plasticity compared to adults[3], the extent to which specific sensory and cognitive traits are fully developed versus those that remain malleable is still debated. For instance, studies on sensory perception have demonstrated that adolescent learning is slower and more variable[4, 5], while others have found that adolescents learn faster in novel contexts, such as reversal learning[6]. Certain traits, such as cognitive flexibility, are more pronounced in adolescents compared to adults, whereas other behaviors such as impulsive control, are still immature[7]. Furthermore, results across studies that target sensory or cognitive traits are often inconsistent[8]. Therefore, understanding the degree to which sensory and cognitive functions have matured and the underlying brain mechanisms that support them remain an ongoing challenge.

A key concept in brain development is that of critical periods (CPs), which are specific time windows during which the brain's neural circuits are highly plastic and particularly sensitive to external experiences[9, 10]. The early postnatal development of sensory cortices provides classical examples of this phenomenon. In mice, neurons in the visual cortex undergo several CPs such as those for orientation selectivity, direction selectivity and ocular dominance. During the CP, which occurs in different developmental time windows for different functional attributes, visual experience shapes the specific neuronal attribute to a long-term state (e.g., ocular dominance reaches its adult form)[9, 11]. The CPs for simple neuronal features in primary sensory cortices are typically complete by the onset of puberty. In the auditory cortex (ACx) of mice, the CP for pure tone processing begins as early as postnatal day 12 (P12), shortly after the ear canals open, and closes by ~P15[12, 13]. Thus, the closure of the auditory CP to pure tones occurs before adolescence begins[14, 15]. Yet, while some studies have found that simple neuronal properties in the ACx (like response properties to pure tones) are nearly mature by adolescence, others report continued maturation well beyond this developmental stage[16, 17]. For example, previous studies found that auditory learning and behavioral performance in auditory discrimination tasks involving amplitude modulation detection, as well as temporal interval discrimination tasks remain highly constrained during adolescence[4, 5]. In those tasks, ACx neurons in adolescents exhibit high neuronal variability and lower tone sensitivity as compared to adults[4].

The perspective shift from viewing development as involving a set of discrete CPs to a broader view of continuous development with multiple, overlapping, CPs that affect each other, calls for studies to test the extent of functional modulation in neuronal features after their CP 'closes'. Thus, although the CP for pure tone representation in the ACx ends by ~P15, we hypothesized that learning of pure tone discrimination would remain malleable in adolescence, because other perceptual features and particularly other cognitive features are still developing. Further, it remains unclear how these features are expressed when measured in the context of active behavior. We measured cortical activity in the ACx of adolescent mice engaged in an auditory discrimination task involving pure tones. Our findings show that learning, behavior, and sound representations to pure tones in the ACx are distinct in adolescence.

Results

Adult mice outperform adolescents on a difficult, but not an easy, auditory discrimination task of pure tones

To study learning and perceptual performance abilities we trained and tested adolescent and adult mice on an auditory pure-tone discrimination Go/No-Go paradigm. For this purpose, we utilized an automated behavioral platform called the ‘Educage’ (Fig. 1a) [18]. We trained adolescent ($n = 15$) and adult ($n = 15$) mice from post-natal day 20 (P20)-P37 and P60-P77, respectively (Fig. 1b). Following a tone association period (3-days; exposure to Go tones when mice enter the drinking port), mice learned to discriminate between two pure tones separated by 1 octave (7.07kHz vs 14.14kHz; Fig. 1b-c, light blue; herein referred to as the ‘easy task’). After one week of training on the easy task (P23 to P30 in pre-adolescent mice and P63 to P70 in adult mice) mice were introduced to a second pure-tone pair, now separated by 0.25 octave (9.2kHz vs 10.9kHz) (Fig. 1b-c, dark blue; herein referred to as the ‘hard task’). We completed the experiment after 1 week of learning on the easy and hard task simultaneously (i.e., at P37 for adolescents and P77 for adults). During the whole period of training, we also presented mice with low-probability, unrewarded ‘probe’ trials of pure-tones spanning the range of learned stimuli (Fig. 1c, grey). Figure 1d shows example learning curves (d') from one adolescent and one adult mouse over the 14-day course of tone discrimination. Adolescents and adults learned the procedure of the task similarly well, as measured by the number of trials it took mice to first reach a discriminability threshold of d' value ≥ 1 (Fig. 1e).

We tested for differences in auditory discrimination by comparing the performance of mice at different epochs along training. First, by the end of the first week of training, all mice performed above the discrimination threshold (Fig. 1f). Although adolescent mice performed significantly fewer trials during the first week of training (Fig. 1f, vertical lines), there was no significant difference between the performance of adult and adolescent mice at the end of the first week (Fig. 1f, horizontal lines; d' calculated from the last 100 trials). Following one week of training on the easy task, we introduced the second pair of pure tones (Fig. 1b; ‘Easy + Hard’). Immediately after introducing the hard pair, the performance of adolescent mice on the easy task dropped, while those of adults remained unaffected (Fig. 1g). During the second week of training adolescents regained their high d' values on the easy task, such that by the end of the experiment both groups had similarly high performance on the easy task (Fig. 1h).

We then compared how mice learned the hard task during the second week while training on both the easy and hard task simultaneously. While both groups now performed a similar number of trials (Fig. 1i, vertical lines), adult mice outperformed adolescent mice on the hard task (Fig. 1i, horizontal lines), suggesting that adolescents struggled with this (harder) level of discrimination. Indeed, on average, adolescent mice did not improve on the hard task during the second week of training (Fig. 1j, adolescents). Adults, on the other hand, improved their performance on the hard task during the second week (Fig. 1j). Notably, neither group could learn the hard task if it was not preceded by the easy task (Fig. S1-1a-b).

Although all mice trained for exactly 14 days, each mouse performed a unique number of trials, as these were initiated by the mice spontaneously. As noted above, pre-adolescent mice performed fewer trials/day as compared to adults but, notably, only during the first week of training (mean $\pm$ SEM: adolescents, 8317 ± 712 trials, adults 13229 ± 514 trials, t -stat = -5.6314, $df = 28$, $p = 4.9566e-06$, two-sample independent t -test). Still, this difference raises the question whether the different number of trials affected our conclusions from the first week of training. Thus, to evaluate the possible connection between the number of trials and performance in our data, we carried out several analyses. First, we found that the number of trials and d' were not correlated in either age-

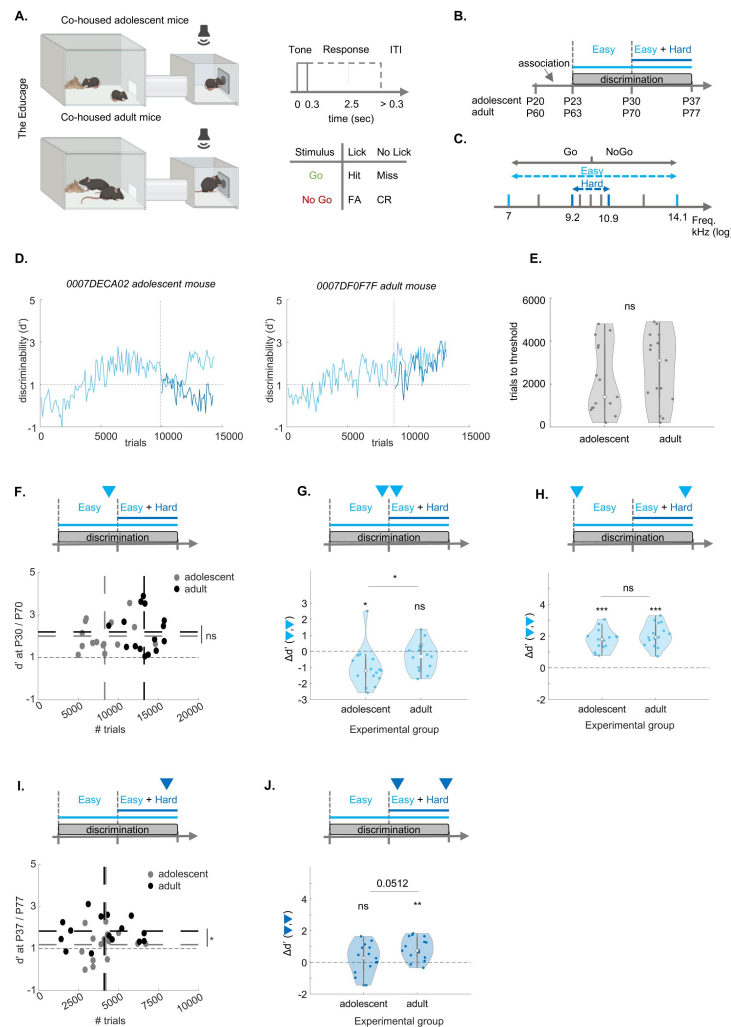


Figure 1

Adolescent mice exhibit lower performance during self-initiated auditory learning in the 'Educage'.

A. Schematic model of the 'Educage' (left), the trial structure and trial types (FA, false alarm; CR, correct reject). Created with *BioRender.com*. **B.** Experimental timeline. Total training time was 21 days (± 2). **C.** The sounds used for training. Light blue-easy task; dark blue-hard task; Grey-probe-trials. **D.** Learning curve examples. Adolescent mouse (left, P20-to-P37); adult mouse (right, P60-to-P77). Vertical dashed lines indicate the easy-hard transition. Horizontal line is $d' = 1$. **E.** Number of trials to reach threshold ($d' \geq 1$); adolescents, $n = 15$; adults, $n = 15$; $z = -0.1659$; $p = 0.8682$, two-sample Wilcoxon rank sum test). **F.** Discriminability (d') of the easy task in adolescent mice at P30 (grey; $n = 15$, $d' = 2.0001 \pm 0.1791$; $n \text{ trials} = 8317 \pm 712$) and adult mice at P70 (black; $n = 15$, $d' = 2.1986 \pm 0.2441$; $n \text{ trials} = 13229 \pm 514$, top: marked by the arrowhead). Dashed lines: mean trials per group ($t\text{-stat} = -5.6314$, $df = 28$, $p = 4.9566e-06$, two-sample independent t -test, vertical line), and mean d' per age group ($z = -0.2074$; $p = 0.8357$, two-sample Wilcoxon rank sum test, horizontal line). **G.** Change in discriminability ($\Delta d'$) of the easy task before and after the introduction of the hard task (Top: arrowheads; left: adolescent, signed rank = 14, $p = 0.0067$; right: adult, signed rank = 14, $p = 0.1514$, one-sample Wilcoxon signed rank test; $\Delta d'$ between adult and adolescent mice: $z = -2.0739$; $p = 0.0381$, two-sample Wilcoxon rank sum test). **H.** Same as 'G' for the first 100 and last 100 trials of the experiment in the easy task (adolescent signed rank = 120, $p = 6.1035e-05$; adult signed rank = 120, $p = 6.1035e-05$, one-sample Wilcoxon signed rank test; $\Delta d'$ between adult and adolescent mice: $z = -1.0370$; $p = 0.2998$, two-sample Wilcoxon rank sum test). **I.** Same as 'F' for the hard task (adolescent-grey; $n = 15$, $d' = 1.1895 \pm 0.1783$; $n \text{ trials} = 4163 \pm 297$; adult-black; $n = 15$, $d' = 1.8342 \pm 0.1743$; $n \text{ trials} = 4102 \pm 475$) (mean trials per group: $t\text{-stat} = 0.1306$, $df = 28$, $p = 0.8970$, two-sample independent t -test, vertical line; mean d' per group: $z = -2.2398$; $p = 0.0251$, two-sample Wilcoxon rank sum test, horizontal line). **J.** Same as 'G' for the hard task. (adolescent signed rank = 73, $p = 0.4887$; adult signed rank = 114, $p = 8.5449e-04$, one-sample Wilcoxon signed rank test; $\Delta d'$ between adult and adolescent mice: $z = -1.9495$; $p = 0.0512$, two-sample Wilcoxon rank sum test).

group during the first week of discrimination on the easy task (Pearson- r ; adolescent: $r = 0.1880$, $p = 0.5023$; adult: $r = -0.0730$, $p = 0.7959$), nor on the second week of discrimination on both the easy and the hard task (Pearson- r ; adolescent: $r = 0.2264$, $p = 0.4171$; adult: $r = 0.0194$, $p = 0.9454$). Second, we compared performance at three time points along the task with reference to times with shared number of trials: 1) evaluating d' after the minimal number of trials performed by all mice during the first week of training, and 2) evaluating d' at the mean number of trials of each group. We found no significant differences between adolescents and adults in all comparisons (Fig. S1-1c-e). Additionally, we found no significant differences between males and females, nor among experiments that had different numbers of co-housed mice in the Educage (Table 1). Taken together, despite the different absolute number of trials during the first week, we conclude that adolescents and adults learned the easy task equally quickly and effectively. The central difference between the age groups was in their ability to learn the hard task during the second week, when adults outperformed adolescents.

Adolescents have higher response bias and behavioral variability

We next asked what perceptual and/or cognitive aspect of behavior is different among the groups. We plotted psychometric curves based on both learned and probe trials from the last 2000 trials of the experiment (Fig. 2a). The false alarm rate but not the hit rate of adolescent mice was significantly higher as compared to adults (Fig. 2b). To calculate the decision boundary of mice and their perceptual sensitivity we normalized the psychometric curves of each mouse using a unity-based normalization and sigmoid fitting (Fig. 2c). We found no differences in the decision boundary or slopes of the curves, suggesting that perceptual sensitivity is not different between the groups (Fig. 2c). To test whether the differences arise from a cognitive effect (i.e., lick bias), we calculated the decision threshold as the maximal lick bias, also known as the criterion bias (C). C reflects the tendency to respond in a liberal (i.e., negative C) or a conservative (i.e., positive C) manner to the sounds during the task. C was negative for both groups, yet significantly lower in adolescents, suggesting that their cognitive ability to withhold licking is inferior to that of adults (Fig. 2d).

Given that adolescents and adults show different dynamics in their performance along the task (Fig. 1), we tested whether the differences we found in the lick bias are consistent at different times along the task. To normalize for different numbers of trials in each mouse, we divided their training episodes into the first and last tertiles during the first week and second week of training (Fig. 2e). The maximal lick bias was significantly higher in adolescents in all but the very first training episode, when mice learned the procedure (Fig. 2e-f). These data highlight the general cognitive difficulty of adolescents to withhold licking during the task. The cognitive sensitivity of adolescents was further evident when their performance dropped right after the task was switched from 'easy only' to 'easy + hard' (Fig. 1f). The lick bias of adolescent mice became significantly stronger after the introduction of the hard task (Fig. 2g). In contrast, the lick bias of adult mice remained stable (Fig. 2g). These data further suggest that this cognitive trait (i.e., to withhold licking) has not yet matured in the young mice. To test if the lick bias was stimulus-related or a result of general impulsivity, we also compared the inter-trial interval (ITI) after different trial outcomes, and observed similar ITIs across groups, except for false alarm trials, where adolescents initiated trials significantly faster (Fig. S2-1). This suggests that adolescent mice exhibit more impulsive responses following punishments, contributing to their immature lick bias.

Prior studies have shown that a dominant feature of adolescent behavior is that it is more variable compared to adults [4]. To examine behavioral variability in our data, we calculated the coefficient of variation (CV) of the discriminability (d') for each mouse across training. While there was no significant difference in d' variability when training on the easy task during the first week of training (Fig. 2h), adult mice showed significantly lower variability during the second week

Fixed effects	Estimate	SEM	T-statistic	DF	P-value**	CI (lower)	CI (upper)
Intercept	1.6001	0.2312	6.9222	27	3.8798e-07	1.1258	2.0744
Age	0.5994	0.2266	2.6457	27	0.0269	0.1346	1.0643
Sex	-0.0428	0.2476	-0.1729	27	0.9999	-0.5509	0.4653

Table 1

Behavioral differences between adolescent and adult mice are age-, but not sex-related.

Fixed effects of age and sex, and the random effects of co-housing in the 'Educage' on the discriminability (mean d' of the last 100 trials of the easy and hard task to avoid pseudo replication) of all mice (Number of observations = 30, Fixed effects coefficients = 3, Random effects coefficients = 7, Covariance parameters = 2). Coefficient estimates, STE, T-statistic, degrees of freedom, p-values (adjusted for multiple comparisons with the Bonferroni method) and the lower and upper Confidence Interval (95%). The model includes random effects coefficients of the Cage ID in each group of co-housed mice (7 cages in total; see *methods*, equation 7). Model structure: $\text{discriminability}(d') \sim \text{age} + \text{sex} + (1 | \text{cage ID})$.

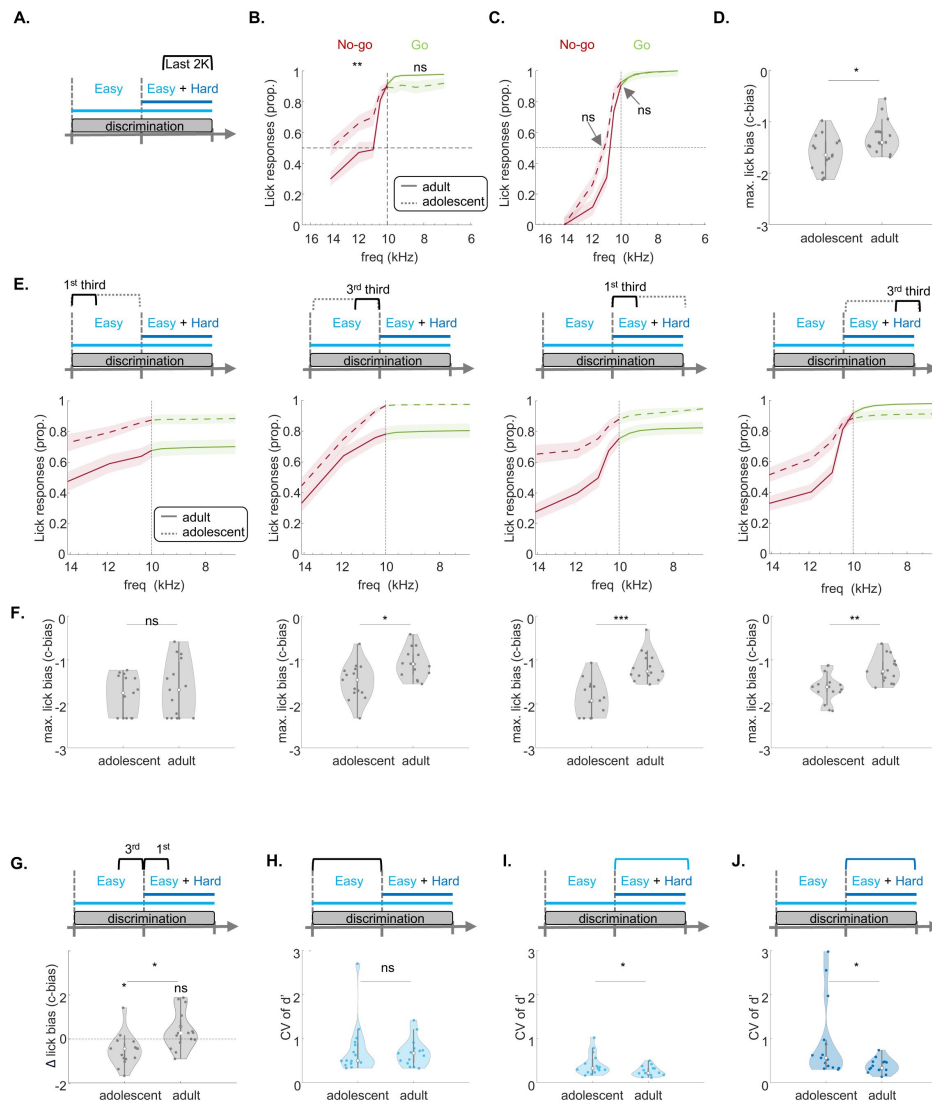


Figure 2

Adolescent mice exhibit lower response inhibition and higher variability in performance over the course of learning.

A. Schematic timeline of tone discrimination). **B.** Average psychometric curve of the proportion of lick responses (above the threshold) across all stimuli (probe and learned; No-Go stimuli: $p = 0.0085$, Go stimuli: $p = 0.0963$, Wilcoxon rank sum test, after Bonferroni correction). **C.** Same as 'B', but normalized and fitted to a sigmoid function (proportion to the lick threshold at 10 kHz, $p = 0.1163$; frequency at 50% proportion to the lick threshold, $p = 0.5896$, Wilcoxon rank sum-test, after Bonferroni correction). **D.** Maximal lick bias (i.e., criterion bias) per mouse during the last 2000 trials (c-bias: $p = 0.0136$, Wilcoxon rank sum test). **E.** Average psychometric curve, fitted to a sigmoid function across all stimuli (probe and learned) at different periods of the experiment — Left: first tertile-easy task (per mouse; adolescent $n = 15$, adult $n = 15$); Middle-left: last tertile-easy task; Middle-right: first tertile-hard task; Right: last tertile-hard task. (Dashed vertical = category boundary). **F.** Same as E. as the maximal C-bias. From left to right: ($z = -0.5104$, $p = 0.6098$, two-sample Wilcoxon rank sum test), ($z = -2.6546$, $p = 0.0079$, two-sample Wilcoxon rank sum test), ($z = -4.0739$, $p = 4.6228 \times 10^{-5}$, two-sample Wilcoxon rank sum test), ($z = -3.2768$, $p = 0.0011$, two-sample Wilcoxon rank sum test). **G.** Change in lick bias (Δ c-bias) between the last 100 trials before and after the introduction of the hard task. (left, adolescent signed rank = 17, $p = 0.0125$; right, adult signed rank = 83, $p = 0.2078$, one-sample Wilcoxon signed rank test; Δ c-bias between adolescents and adults, $z = -2.9035$, $p = 0.0037$, two-sample Wilcoxon rank sum test). **H.** Coefficient of variation (CV) of the d' of the easy task before the introduction of the hard task. ($z = -0.1659$, $p = 0.8682$, two-sample Wilcoxon rank sum test). **I.** Same as 'H' after the introduction of the hard task ($z = 2.1569$, $p = 0.0310$, two-sample Wilcoxon rank sum test). **J.** Same as 'H' but for d' of the hard task ($z = 2.5302$, $p = 0.0114$, two-sample Wilcoxon rank sum test).

of training while they trained on both the easy and hard tasks (Fig. 2i, j). Thus, adolescent learning and behavior are characterized by both lower response inhibition and higher variability, particularly during the second week when the task involved more challenging discrimination.

Adult mice outperform adolescent mice on a head-fixed discrimination task

To enable electrophysiological recordings during expert task performance, we trained adolescent ($n = 5$) and adult ($n = 6$) mice on a head-fixed learning paradigm, using a similar protocol as in the Educage (Fig. 3a). In the head-fixed protocol, water access was limited to training sessions. Head-fixed mice were trained to lick after a 100ms tone and were rewarded (with water) or punished (by a 2 sec white noise) after a 600ms delay (Fig. 3b). Each session was concluded after the mice became satiated and stopped licking. Water supply was limited to 0.0125 ml per day per gram of body weight, and mice that did not consume this amount were compensated after training (see Methods) (Fig. 3c shows an example session from one mouse). The number of trials per training session was significantly lower in adolescents (mean $\pm$ SEM: adolescents: 410 ± 35 , adults: 580 ± 37 t-statistic = -27.9855, $p = 0.0012$, independent two-sample t-test). However, the number of sessions to reach the behavioral threshold ($d' \geq 1$ in the easy task) was not significantly different between adolescents and adults (mean $\pm$ SEM: adolescents: 3.7 ± 0.8 , adults: 3.8 ± 0.7 ; t-statistic = -0.8531, $p = 0.1971$, independent two-sample t-test, Fig. 3d). While the number of trials performed within each session is likely related to physiological factors, such as smaller body weight and faster satiation, they did not impact learning capabilities. Similar to learning in the Educage, adolescent performance (but not that of adults) decreased after we introduced the hard task (Fig. 3e). Another similarity between the head-fixed and Educage versions of the task was that we found no significant differences between adolescents and adults in the performance of the easy task (Fig. 3f), and that adults outperformed adolescents on the hard sound discrimination (Fig. 3g). Also similar to the behavior in the Educage, lick rates of adolescent mice were generally higher (Fig. S3-1a), and adolescents exhibited a stronger lick bias (Fig. S3-1b). In contrast, in the head-fixed task we found no evidence in impulsiveness; there were no differences between the groups in the licking patterns during the inter-trial intervals (Fig. S3-1c, d).

To test whether ACx was necessary for expert discrimination in this task, we performed a causal experiment in adult mice. Adult mice ($n=3$) were injected bilaterally with AAV5-CAMKII-GtACR2-FRED_kv_2.1 into the ACx, and optical fibers were implanted over the injection sites (Fig. S3-2a,c; Fig. S3-3). This viral construct enables optogenetic silencing of excitatory neurons via GtACR2 (*Guillardia theta* Anion Channelrhodopsin 2), under the control of the CaMKII promoter. It includes a RED Fluorescent protein for expression verification (FRED) and a soma-targeting motif derived from the Kv2.1 potassium channel to enhance membrane localization. Mice were trained identically on the head-fixed Go/No-Go paradigm as described above. At the end of training, we attached a light source to the implanted fibers to allow optogenetic suppression during several sessions (Fig. S3-2a). In these sessions, light was applied in 50% of trials in a pseudo-random fashion, with light ON trials starting from -50ms prior to tone onset up to 50ms after tone offset (Fig. S3-2b). A control group of mice ($n = 3$) were injected with AAV9-CaMKII-dTomato and went through the exact same procedure (Fig. S3-2a,c). We compared the lick rates for both Go and No-Go stimuli under light ON and light OFF conditions. We found that lick rates were strongly affected in the GtACR2-injected mice ($n = 13$ sessions), but not in controls ($n = 8$ sessions) (Fig. S3-2d). Light ON conditions only affected the lick rate of the inhibited trial and not the follow-up trials (Fig. S3-2e). Discriminability (d') decreased significantly with optogenetic suppression for both easy and hard tasks, although mice still performed better in the easy task under both conditions (Fig. S3-2f). These results confirm findings by others in similar (though not identical) tasks [19, 20], suggesting that the ACx is necessary for executing the behavior during the expert stage of this task (but see Discussion).

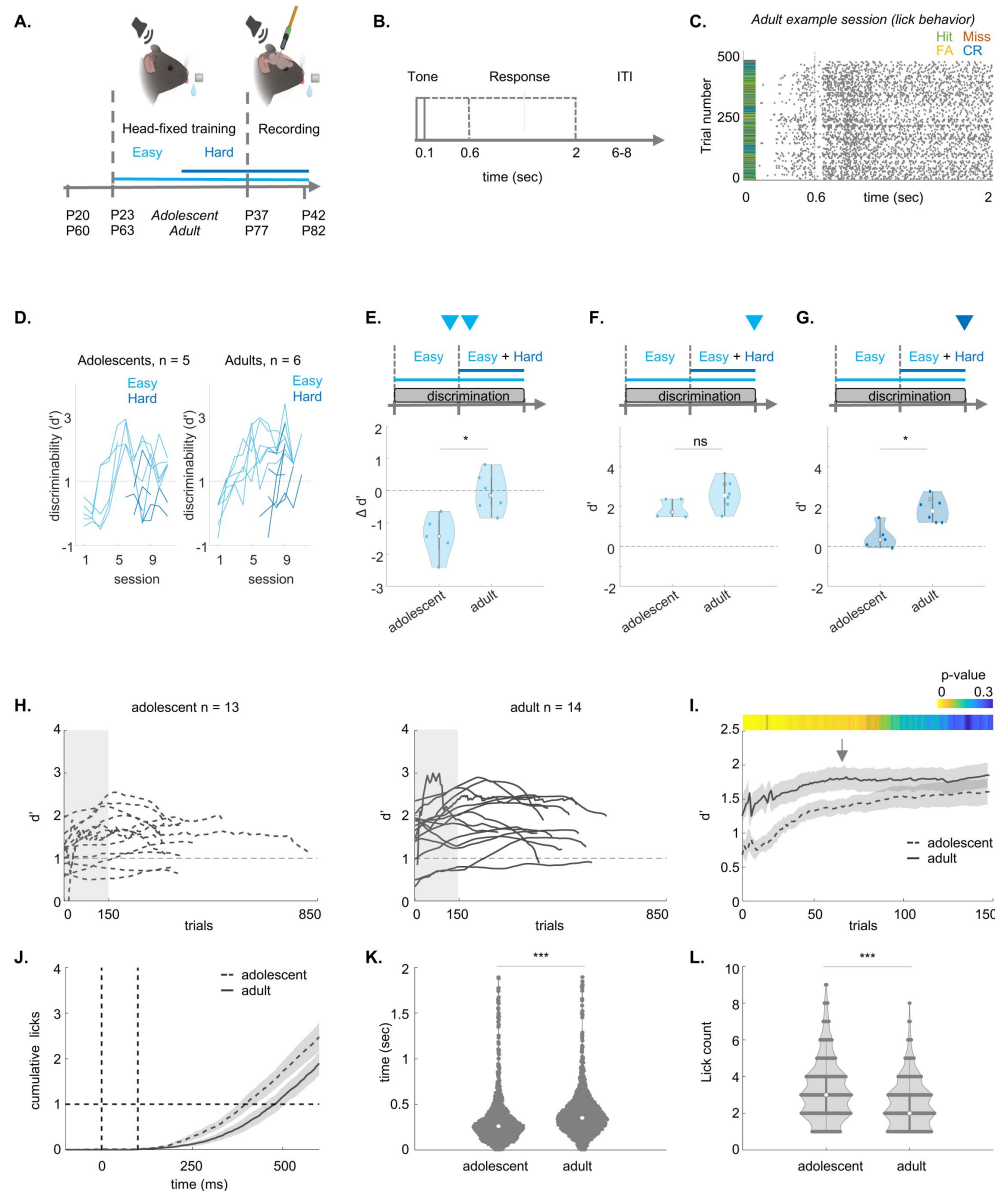


Figure 3

Adolescent mice exhibit lower performance in the head-fixed discrimination task.

A. Experimental timeline of training followed by recordings. Created with *BioRender.com*. **B.** Trial structure during the recording. Solid lines indicate the tone period. Dashed lines show the reward or punishment delay (0.6 sec), and the response window (2 sec). **C.** Example session. Licks (grey ticks) and trial outcomes (hit = green, false alarm = yellow, miss = red and correct reject = blue) across all trials in one recording session. **D.** Discriminability during training sessions for the easy task (light blue) and hard task (dark blue). **E.** Change in d' after the introduction of the hard task (last 100 trials of the last session of the easy task compared to last 100 trials of the first hard session; rank-sum = 14; $p = 0.0381$, two-sample Wilcoxon rank sum test). **F.** Expert d' of the last 100 trials during the last training session of the easy task (rank-sum = 21; $p = 0.1255$, two-sample Wilcoxon rank sum test). **G.** Same as 'F', but for the hard task (rank-sum = 17; $p = 0.0173$, two-sample Wilcoxon rank sum test). **H.** Behavioral performance (average d' of the easy and the hard task) per mouse during recording sessions for adolescents ($n = 13$, left) and adults ($n = 14$, right) (trials per recording: adolescent: 340.5385 ± 45.0650 ; adult: 431.1429 ± 30.3367 ; independent t-test, t-statistic = -203.7581, $p = 0.1116$). **I.** Same as 'H' but only for the first 148 trials. The color bar shows the p-values between the groups. **J.** Average cumulative licks per trial in adolescents (dashed-line) and adults (solid-line) from -200ms before tone - onset until the reward or punishment delay, 500ms after tone-offset. **K.** Lick latency per trial for adolescent (left) and adult (right) groups during electrophysiological recordings (LME statistics are shown in supplemental Table 1). **L.** Same as 'K' for the Lick count.

To measure neural responses during behavioral performance, head-fixed expert mice underwent multiple recording sessions targeting the ACx (adults, $n = 14$ recording sessions from 6 mice, age: P77-P82; adolescents, $n = 13$ recordings from 5 mice, age: P37-P42). During the recordings, the performance of all mice on the easy task was at least $d' > 1$, and the d' on the hard task was smaller for both groups (often smaller than 1) (Fig. S3-4a). Performance was not significantly different across the multiple recording sessions in each mouse (Fig. S3-4b). Behavioral performance during recordings was not significantly different to performance in the Educage (Fig. S3-4c). **Figure 3h** shows the evolution of discriminability (d') during all recordings for adolescents and adults. The performance was heterogeneous across mice as well as within recording-sessions. However, the total number of trials performed was not significantly different (trials per recording: adolescent: 340.5 ± 45 ; adult: 431.1 ± 30.3 ; independent t-test, t-statistic = -203.7581, $p = 0.1116$).

We found additional differences between the behavior of adults and adolescents in the head restrained configuration. The average adolescent discriminability was lower at the beginning of the session (initial 78 trials) but then rapidly leveled out (**Fig. 3i**). Adolescent mice had shorter lick latencies (**Fig. 3j, k**), and higher lick counts (**Fig. 3j, l**; Supplementary Table 1). While the lick latencies were independent of the discriminability (d'), we found a significant interaction effect between lick count and the d' (Supplementary Table 1). Together, these behavioral differences suggest that the weaker response inhibition (here, expressed as lick latency) and higher reward anticipation (here, expressed as lick count) may contribute to the weaker performance of adolescents during the beginning of the recording.

Neuronal representations of stimulus- and choice-related activity are immature in adolescents

To compare the activity of neurons in adolescent and adult mice, we recorded spiking activity from the ACx using the high-density Neuropixels-1 probes in expert mice engaged in the task. We targeted the ACx by inserting the probe in a diagonal angle, traversing four auditory regions: Dorsal Auditory Cortex (AUDd), Primary Auditory Cortex (AUDp), ventral auditory cortex (AUDv), and Temporal Association Cortex (TEa) (**Fig. 4a** left, Fig. S4-1a-b show an example voltage trace and one example recording). We recorded multiple times from each mouse and used DiI- or DiO-coatings for different penetrations to verify probe trajectories postmortem. We used the 3D-allen CCF-slice reconstruction (github.com/cortex-lab/allenCCF), for high resolution anatomical registration (**Fig. 4a, b**; Fig. S4-1c, d).

We limited our single neuron analysis to well-isolated single units from infragranular layer 5 and layer 6 (recordings were restricted to these layers due to the probe angle), as two main projection layers of the cortex, which are key nodes for assessing cortical outputs [21]. Recorded neurons from other layers and areas were excluded from the analysis. Together, we collected data from 1145 single neurons in 13 recordings of 5 adolescent mice and 1267 single neurons in 14 recordings of 6 adult mice (Supplementary Table 2; an overview of every dataset can be found in Supplementary Table 7). We further restricted our analysis to single neurons excited by sounds during the first 150ms after tone onset, to exclude any contributions of motor actions due to licking (**Fig. 4c**; Fig. S4-1e). Lick rate- and Firing rate-PSTHs were not correlated during the first 150ms (adolescents $r = -0.1515$, $p = 0.9878$; adults $r = -0.0638$, $p = 0.9942$). In total, we analyzed $n = 463$ neurons from adolescent mice and $n = 599$ from adults.

In **Table 2**, we present the number of neurons in our dataset, categorized by region. A detailed overview of the differences in firing properties per auditory region is also provided in Figure S4-2 and Supplementary Table 3. Consistent with previous work, we found that firing properties such as minimal latency differed between auditory regions in both age groups, with AUDp and TEa being most distinct (Fig. S4-2, Supplementary Table 3) [22].

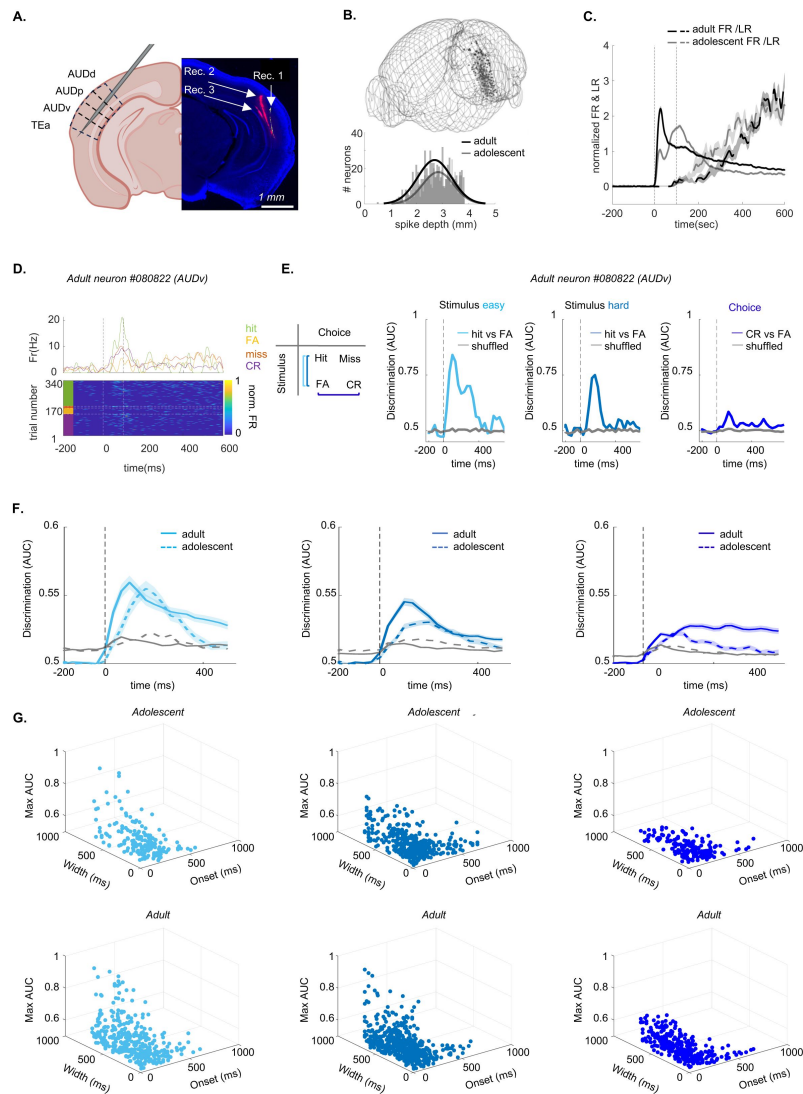


Figure 4

ACx neurons in adolescents exhibit lower discriminability in stimulus- and choice-related activity

A. Recordings in ACx when the mouse is engaged in the task, using Neuropixels-1 probes. Left: Recordings were performed in AUDd, AUDp, AUDv, and TEa. Right: Fluorescent micrograph of a coronal brain slice showing the probe tracks of three recordings (red = DiI, yellow = DiO). Created with *BioRender.com*. **B.** Top: 3D-Reconstruction of recording sites in adolescents ($n = 13$; grey) and adults ($n = 14$; black). Bottom: distribution of the spike-depth of all excitatory tone-responsive L5/6 neurons in adolescents ($n = 455$; grey) and adults ($n = 607$; black). **C.** Normalized PSTH and lick rate (LR) from -200ms to +600ms after tone-onset in adolescents (grey) and adults (black). **D.** Spiking activity from one example neuron sorted by trial outcome (hit, miss, false alarm, correct reject). Top: PSTH per trial outcome. Bottom: Heat map of the FR sorted per trial outcome. **E.** Discriminability values (AUC) over time (from -200ms to 600ms after tone onset) for one example neuron (same neuron as in 'D'). AUC values are shown for stimulus related activity (left: easy task, middle: hard task) and choice-related activity (right). Shuffled distribution in all curves is shown in grey. **F.** Same as 'E' for all neurons. The curves are average ($\pm$ SEM) neuronal discriminability of adult neurons (solid line) and adolescent neurons (dashed line), for easy (adolescent neurons = 190, mice = 4, recordings = 7; adult $n = 358$, mice = 4, recordings = 8; left) and hard stimulus-related activity (adolescent $n = 429$; adult $n = 562$, mice = 5, recordings = 9; middle), and choice-related activity (adolescent $n = 429$; adult $n = 562$, mice = 5, recordings = 9; right). **G.** 3D plots of the onset-latency of discriminability (ms), duration of discriminability (ms), and maximal discriminability (AUC) of all neurons that showed significant discriminability. Left: easy task (adolescent neurons = 178 (93%), mice = 4, recordings = 6; adult $n = 346$ (97%), mice = 4, recordings = 8; left); Center: hard task (adolescent neurons = 399 (93%), mice = 5, recordings = 10; adult $n = 544$ (97%), mice = 6, recordings = 12; middle); Right: choice-related activity (adolescent neurons = 181 (95%), mice = 4, recordings = 9; adult $n = 339$ (95%), mice = 4, recordings = 7; right).

	Stimulus easy						
Fixed Effects	Estimate	StandardError	tStatistic	DF	pValue	CI lower	CI upper
	Intercept	163.6104	7.9457	20.591	522	6.29587E-69	148.0009 179.2199
	Onset (ms)	-29.7978	9.7626	-3.0523	522	0.00716101	-48.9765 -10.6191
	Intercept	216.2921	13.2858	16.2799	522	5.15393E-48	190.1919 242.3924
	Duration (ms)	63.2599	16.3499	3.8691	522	0.000369146	31.1401 95.3796
	Intercept	0.6035	0.007	85.6404	522	0.0001	0.5897 0.6174
	Max AUC	0.0172	0.0087	1.9776	522	0.145507056	0.0001 0.0342
	Stimulus hard						
Fixed Effects	Estimate	StandardError	tStatistic	DF	pValue	CI lower	CI upper
	Intercept	151.1905	5.4684	27.6479	941	3.6685E-123	140.4588 161.9222
	Onset (ms)	-35.3357	7.1998	-4.9079	941	3.25392E-06	-49.4651 -21.2063
	Intercept	220.614	8.4357	26.1525	941	2.8344E-113	204.0591 237.1689
	Duration (ms)	55.6268	11.1065	5.0085	941	1.9647E-06	33.8305 77.4231
	Intercept	0.5821	0.0034	170.2777	941	0.0001	0.5754 0.5888
	Max AUC	0.0098	0.0045	2.1862	941	0.087143965	0.001 0.0187
	Choice						
Fixed Effects	Estimate	StandardError	tStatistic	DF	pValue	CI lower	CI upper
	Intercept	174.5856	8.259	21.139	518	1.6732E-71	158.3605 190.8108
	Onset (ms)	-24.3644	10.2288	-2.3819	518	0.052746707	-44.4595 -4.2693
	Intercept	228.1768	12.92	17.6607	518	1.64628E-54	202.7947 253.5589
	Duration (ms)	82.0002	16.0017	5.1245	518	1.26556E-06	50.564 113.4363
	Intercept	0.5491	0.0027	200.1358	518	0.0001	0.5437 0.5545
	Max AUC	0.0166	0.0034	4.8959	518	3.92843E-06	0.01 0.0233

Table 2

Neuronal discrimination is later, shorter, and less precise in adolescent neurons.

Linear mixed effect models of the neuronal discriminability in adolescence and adulthood per stimulus-related activity in the easy task (Number of observations = 524, Fixed effects coefficients = 2, Random effects coefficients = 10, Covariance parameters = 3), stimulus related activity in the hard task (Number of observations = 943, Fixed effects coefficients = 2, Random effects coefficients = 14), and choice-related activity (Number of observations = 520, Fixed effects coefficients = 2, Random effects coefficients = 10, Covariance parameters = 3). The table shows the fixed effects of the coefficient estimates, STE, T-statistic, degrees of freedom, p-values (corrected for multiple comparisons with Bonferroni-correction) and the upper and lower CI of the effect of age on the onset latency of discrimination, duration of discrimination and maximal neuronal discrimination (AUC). Each model also included random effect coefficients of each mouse, and recording per mouse. P-values for were adjusted with post-hoc tests using Bonferroni-correction (see *methods*, equation 9). Model structures: onset latency (ms) ~ age + (1 | Mouse ID) + (1 | Recording ID); duration (ms) ~ age + (1 | Mouse ID) + (1 | Recording ID); maximal discriminability (AUC) ~ age + (1 | Mouse ID) + (1 | Recording ID).

Notably, the effect size of the differences in baseline firing rates (FR) between groups was larger in AUDd, AUDp, and AUDv. However, the evoked FR was smaller in all four areas. The coefficient of variance of the FR was larger in AUDd, AUDp, and AUDv. Additionally, the latency to peak was larger in AUDd and AUDp, while the full-width half-maximum of the FR was also larger in these two areas. The minimal latency was larger in all four areas, and the number of responsive trials was smaller in all four areas. Finally, the lifetime sparseness (see Methods) was smaller in AUDp and AUDv between adolescent and adult neurons (Supplementary Table 3).

Next, to study how adolescents and adults encode task performance we divided neuronal responses by trial outcome (**Fig. 4d**). Specifically, we assessed neuronal discriminability between stimulus-related and choice-related activity by calculating the area under the curve (AUC) from a receiver operating curve (ROC) (**Fig. 4e**). Stimulus-related activity was calculated as the difference between hit and false alarm trials in the easy and the hard task separately. Choice-related activity was calculated as the average difference between false-alarm and correct reject trials (miss trials were excluded since we had insufficient trial numbers of this outcome). **Figure 4e** shows the average AUC values over time from an example neuron (same neurons shown in **Fig. 4D**), and **figure 4f** for all neurons. The vast majority of neurons successfully discriminated (AUC above shuffled data) either or both stimulus- and choice-related activity (adolescent: easy stimuli = 93%, hard stimuli = 93%, choice = 95%; adult: easy stimuli = 97%, hard stimuli = 97%, choice = 95%). The onset-latency of discriminability was significantly slower, maximal discriminability significantly weaker, and the duration of discriminability significantly shorter in adolescent neurons compared to adult neurons (**Table 2**; plotted in **Fig. 4g**). We found no differences between auditory regions (Fig. S4-3). Thus, auditory processing of stimulus- and choice-related activity in the adolescent ACx has not yet reached full maturity.

Population decoding of the hard task in adults outperforms adolescents

While individual neurons exhibited weaker discriminability in adolescents than in adults, at a population level these effects could saturate out. We therefore tested the ability of recorded populations to decode trial outcomes. We used a linear discriminant analysis (LDA) decoder to quantify hit versus correct reject trial outcomes. This comparison extracts information of the combination of stimulus and choice (i.e. task-related differences). We decoded separately in easy trials and hard trials, based on population activity during the first 200 milliseconds. For the easy task there was no significant difference between decoding accuracy in adolescents and adults (**Fig. 5a**). However, decoding accuracy was significantly higher in adults compared to adolescents in hard trials (**Fig. 5a**). For both adults and adolescents, decoding accuracy was significantly lower in hard trials compared to easy trials (**Fig. 5a**). We performed finer timescale decoding to quantify the onset latency of differences between the activity for hit vs. correct reject trials, defined as the time in which the decoding accuracy first crosses three times the standard deviation away from the baseline (see Methods). We found that the latency was shorter in adults in both the easy and hard trials (**Fig. 5b**). In adolescent animals the latency was larger in hard trials (**Fig. 5b**) but not in adult animals (**Fig. 5b**). We then extended this analysis over a temporal window from -0.5s to 10s relative to tone onset (**Fig. 5c**). In the 200ms window after the response period (2.0-2.2 s), there is no longer a significant difference in decoding accuracy between adult and adolescent mice as well as between the two tasks (Fig. S5-1).

To further quantify the different decoding performance between two tasks, we plotted the fisher separation metric ($\frac{\sigma_{between}^2}{\sigma_{within}}$ see Methods) for each level of difficulty. The separation ratio of hard over easy tasks was significantly higher in adult mice than in adolescent mice (robust linear regression, see Methods). Separation could have changed due to alterations in the mean or the dispersion around the mean. Testing the dispersion, we found that the variance around the means

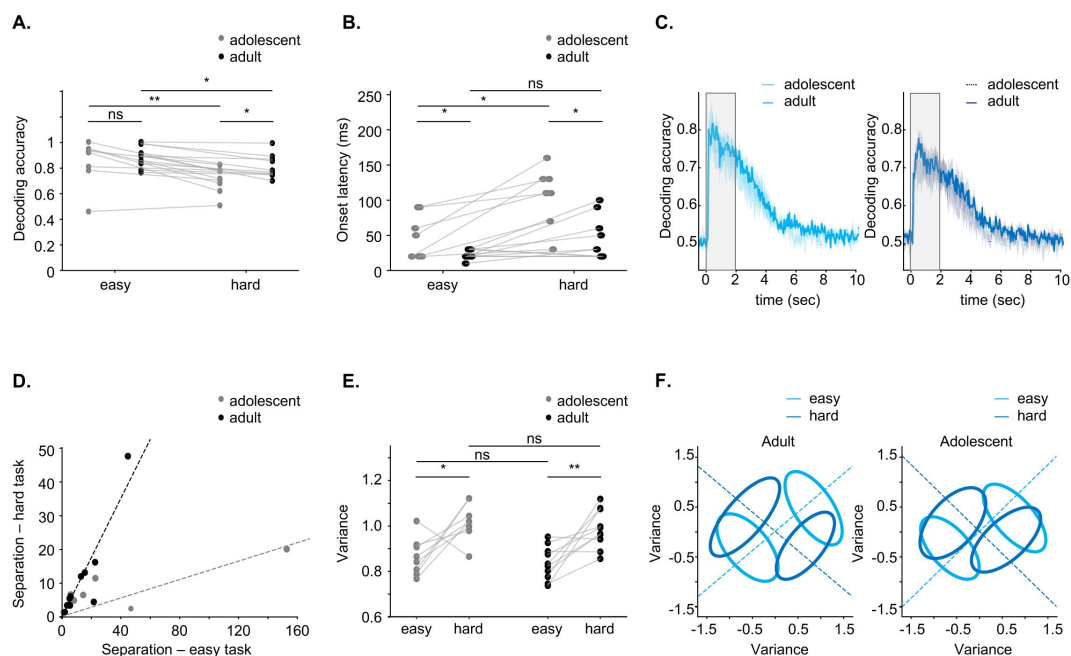


Figure 5

Decoding in adult neuronal populations outperforms decoding in adolescents.

A. Decoding accuracy for the first 200ms across all recordings in both adults (black) and adolescents (grey) for the easy task (adolescents compared to adults, $p=0.5000$, Student's t-test) and the hard task (adolescents compared to adults, $p=0.0300$, Student's t-test). Decoding is better in the easy task for both age groups (adults: $p=0.0030$; adolescents: $p=0.01$, paired t-test). **B.** Decoding latency for all recordings in the easy task ($p=0.0200$, Student's t-test) and the hard task ($p=0.0030$, Student's t-test), as well as compared between age groups (easy task, $p=0.05400$, paired t-test; hard task: $p=0.0100$). **C.** Decoding accuracy over a time window from -0.5s to 10s (the response window highlighted in the grey) for the easy task (left) and the hard task (right). **D.** LDA separation for easy and hard tasks. Lines represent robust linear regression fits without intercept (Huber loss; robust linear regression, $p=0.0001$). **E.** Single trial variance for easy and hard tasks in adolescent and adult recordings (adults: $p=0.0040$; adolescents: $p=0.0300$, paired t-test; easy task: $p=0.4500$; hard task: $p=0.4100$, Student's t-test). **F.** Visualization of population representations for the stimuli in easy and hard tasks. Dotted lines indicate decoding dimensions, and ellipses represent the covariance of the representations.

was not equal across easy and hard trials but rather significantly increased for the hard trials (**Fig. 5d-f**). However, there was no significant difference in variance between adolescent and adult mice within the same task (**Fig. 5d-f**).

Effects of age and learning on cortical plasticity in mice engaged in the task

We next studied how learning contributes to cortical plasticity in the different age groups. To separate between age- and learning-related effects we recorded from the ACx of two new groups of mice (adolescents and adults) that are novice on the task (**Fig. 6a**, P38, recordings= 6; P78, recordings =6, n = 3 mice per group). We collected 130 tone-responsive (by excitation) neurons in novice adolescents and 186 tone-responsive (by excitation) neurons in novice adults (**Fig. 6b-c**; Supplementary Table 2). Notably, novice mice did not discriminate between sounds; instead, they licked indiscriminately to the tones presented during the task (**Fig. S61a-b**).

To reveal learning-related differences, we compared novice and expert mice of similar ages. We computed single-neuron AUCs for Go and No-Go trials (here we considered all trial outcomes to reliably assess differences between novice and expert mice, since expert mice mostly performed CR and novice mice mostly performed FA to No-Go tones), in both easy and hard stimulus pairs (**Fig. 6d-g**). The onset-latency for AUC discrimination was significantly different between adolescents and adults, but not between novice and expert mice, nor between the easy and the hard task. The maximal AUC discriminability was higher, and the duration longer in adults compared to adolescents, between experts and novice, and between the easy and hard tasks (**Table 3**). These results suggest that plasticity is reflected in the interaction of age and learning.

To compare neuronal discriminability and behavior more explicitly, we analyzed the correlations between neuronal decoding and behavioral performance using linear regression. In the easy task, both age groups showed high correlation between neuronal discriminability and behavioral discriminability (**Fig. 6h-i**; light blue). In the hard task, only adult neurons had significant correlations between neuronal discriminability and behavioral discriminability (**Fig. 6h-i**; dark blue).

Effects of age and learning on tuning properties in passively listening mice

Auditory discrimination learning has been shown to induce changes in the tuning properties of sounds in the ACx [18, 19, 21, 23–28]. To study whether learning-induced plasticity in ACx is distinct in adolescents and adults, we recorded from the same mice described above under passive listening condition, by simply extending the recording following the engaged session (**Fig. 7a**, Supplementary Table 5; Adolescents recordings = 4; Adults recordings = 4). These data were compared to a group of novice mice, recorded under passive listening following their engaged session (Supplementary Table 5; adolescent recordings =6; adult recordings =6).

We characterized the response profile of ACx neurons using a ‘frequency-response area’ protocol composed of twenty pure tones (4–40 kHz, spaced at 0.1661 octaves steps), each played for 100ms at five different attenuations (72–42 dB SPL). We limited our analysis to significant excitatory units only, determined as being auditory responsive in a window from tone onset to 50ms after tone offset at 62 dB SPL. Our dataset included four groups. The two adolescent groups were: 1) adolescent novice, and 2) adolescent experts, with 92 and 80 neurons, respectively. The two adult groups were: 3) adult novices, and 4) adult experts, with 123 and 84 neurons, respectively. Neurons in all groups responded to pure tone as expected from previous studies in adult mice, including classical V-shaped frequency response areas [22, 29–32]. A representative example from a responsive neuron of the adolescent expert group and one from the adult expert group are shown in **Figure 7b-c**. We next compared the firing properties of different sub-

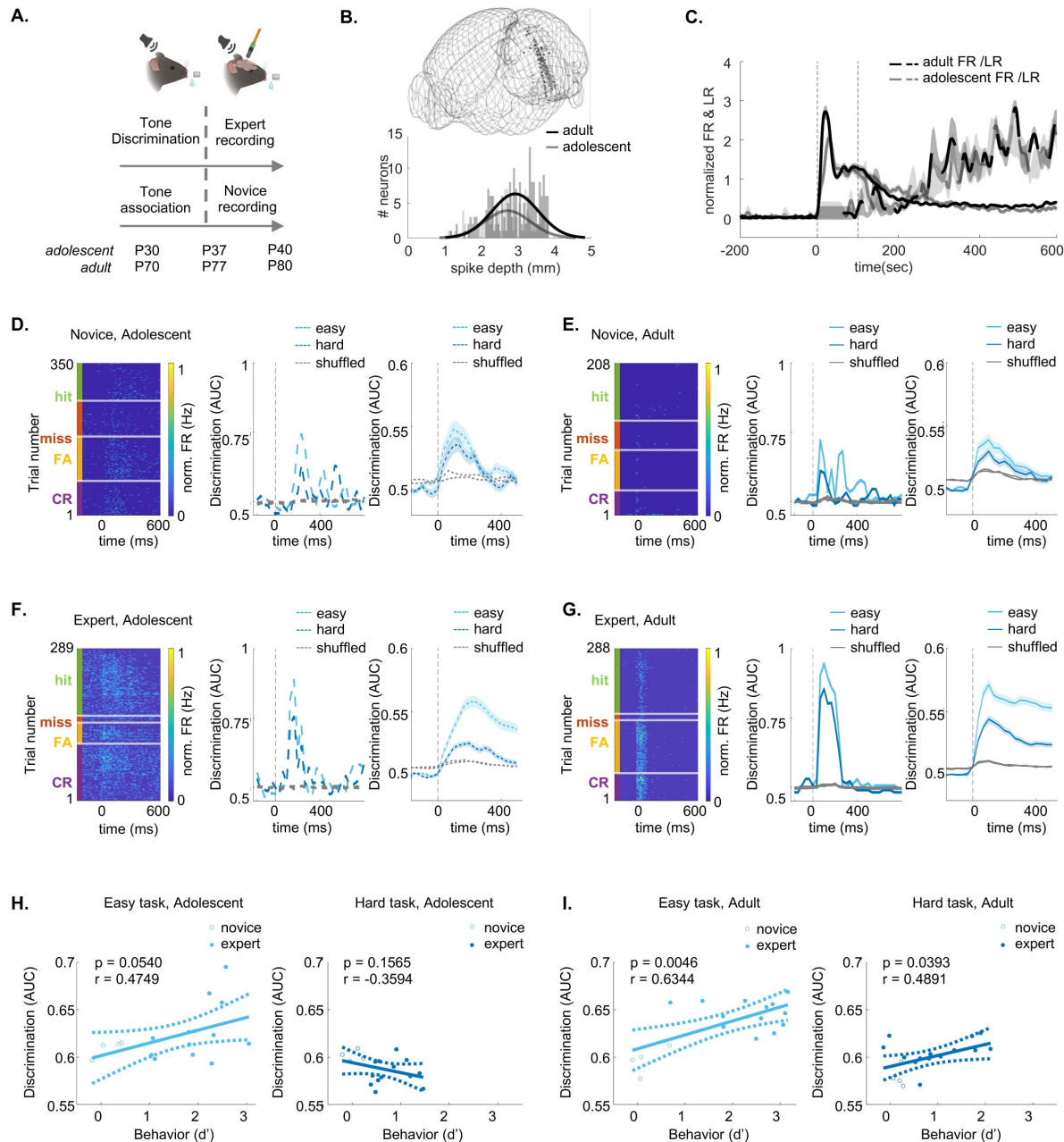


Figure 6

Cortical activity during behavior reflects both age- and learning-induced effects.

A. Training and recording schedule for novice mice, compared to expert mice. Created with [BioRender.com](https://www.biorender.com). **B.** 3D-Reconstruction of recording sites in novice adolescent (n = 6; grey) and novice adult (n = 6; black) mice. Bottom: spike-depth of excitatory tone-responsive L5/6 adolescent (n = 107; grey) and adult (n = 177; black) neurons. **C.** Normalized FR and lick rate (LR) PSTH from -200ms to 600ms after tone-onset in adolescents (grey) and adults (black). Average $\pm$ sem. **D.** Single neuron data from novice adolescent mice. Left: Heat map of the FR per trial from one example neuron sorted by trial outcomes. Center: the AUC of the neuron from the left for the easy and hard stimulus pairs (light and dark blue, respectively). Right: Average ($\pm$ SEM) AUC of all neurons in the novice group (n = 140 neurons). **E-G.** Same as 'D' for novice adult (n = 186 neurons), expert adolescents (n = 455 neurons; Easy vs hard), and expert adults (n = 604 neurons; Easy vs hard). **H.** Linear regression analysis between the average AUC per recording and the behavioral d' during the recording (the correlation and p values are indicated for each plot). **I.** Same as 'H' for adult mice.

Fixed Effects Onset Latency(ms)	Estimate	SEM	T-Statistic	DF	P-Value	CI lower	CI upper
Intercept	92.6519	3.8909	23.8125	2582	5.4024E-113	85.0223	100.2815
Age	-22.6378	5.0624	-4.4718	2582	2.42782E-05	-32.5645	-12.7111
Learning	5.6206	8.7235	0.6443	2582	0.9999	-11.4851	22.7263
Task difficulty	0.8813	5.4571	0.1615	2582	0.9999	-9.8194	11.5819
Age- Learning	-6.8831	11.1114	-0.6195	2582	0.9999	-28.6713	14.905
Age - Difficulty	0.2461	7.2185	0.0341	2582	0.9999	-13.9085	14.4006
Learning - Difficulty	-2.5145	12.3257	-0.204	2582	0.9999	-26.6837	21.6546
Age-Learning- Difficulty	4.4526	15.7691	0.2824	2582	0.9999	-26.4688	35.374
Fixed Effects Duration of Discrimination (ms)	Estimate	SEM	T-Statistic	DF	P-Value	CI lower	CI upper
Intercept	299.7098	7.8305	38.2747	2582	8.4469E-254	284.3551	315.0645
Age	63.2136	9.2513	6.833	2582	3.1017E-11	45.0729	81.3544
Learning	-88.8699	15.9421	-5.5745	2582	8.21813E-08	-120.1306	-57.6092
Task difficulty	-53.9606	9.9745	-5.4098	2582	2.06623E-07	-73.5194	-34.4017
Age- Learning	-30.6998	20.3155	-1.5112	2582	0.3926	-70.5361	9.1365
Age - Difficulty	-18.8772	13.194	-1.4307	2582	0.4579	-44.749	6.9947
Learning - Difficulty	53.8178	22.5289	2.3888	2582	0.0509	9.6414	97.9943
Age-Learning- Difficulty	12.9698	28.823	0.45	2582	0.9999	-43.5488	69.4884
Fixed Effects Maximal Discrimination (AUC)	Estimate	SEM	T-Statistic	DF	P-Value	CI lower	CI upper
Intercept	0.6278	0.0042	148.636	2582	0.0001	0.6195	0.6361
Age	0.024	0.0056	4.2953	2582	5.42797E-05	0.0131	0.035
Learning	-0.0249	0.0096	-2.5859	2582	0.029303232	-0.0438	-0.006
Task difficulty	-0.0438	0.006	-7.24	2582	1.76807E-12	-0.0556	-0.0319
Age- Learning	-0.03	0.0123	-2.4376	2582	0.0446	-0.0541	-0.0059
Age - Difficulty	-0.0023	0.008	-0.2846	2582	0.9999	-0.0179	0.0134
Learning - Difficulty	0.0394	0.0136	2.8893	2582	0.0116	0.0127	0.0662
Age-Learning- Difficulty	-0.0094	0.0175	-0.5387	2582	0.9999	-0.0437	0.0248

Table 3

The effect of age, learning and task difficulty on the latency, duration, and ability to discriminate tones in ACx neurons.

Linear mixed effect models of the effect of age, learning and task difficulty on onset-latency of discrimination, duration of discrimination and maximal discriminability (Number of observations = 2590, Fixed effects coefficients = 8, Random effects coefficients = 20, Covariance parameters = 3). The table shows the fixed effects of the coefficient estimates, STE, T-statistic, degrees of freedom, p-values (corrected for multiple comparisons with Bonferroni-correction) and the upper and lower CI. The model also includes random effects coefficients of each mouse (adolescent novice = 3, adult novice = 3, adolescent expert = 5, adult expert = 6) and recording per mouse (n = 3). P-values for were adjusted with post-hoc tests using Bonferroni-correction (see *methods*, equation 10). Model structures: onset latency (ms) ~ age* learning * difficulty + (1 | Mouse ID) + (1 | Recording ID); duration (ms) ~ age* learning * difficulty + (1 | Mouse ID) + (1 | Recording ID); maximal discriminability (AUC) ~ age* learning * difficulty + (1 | Mouse ID) + (1 | Recording ID).

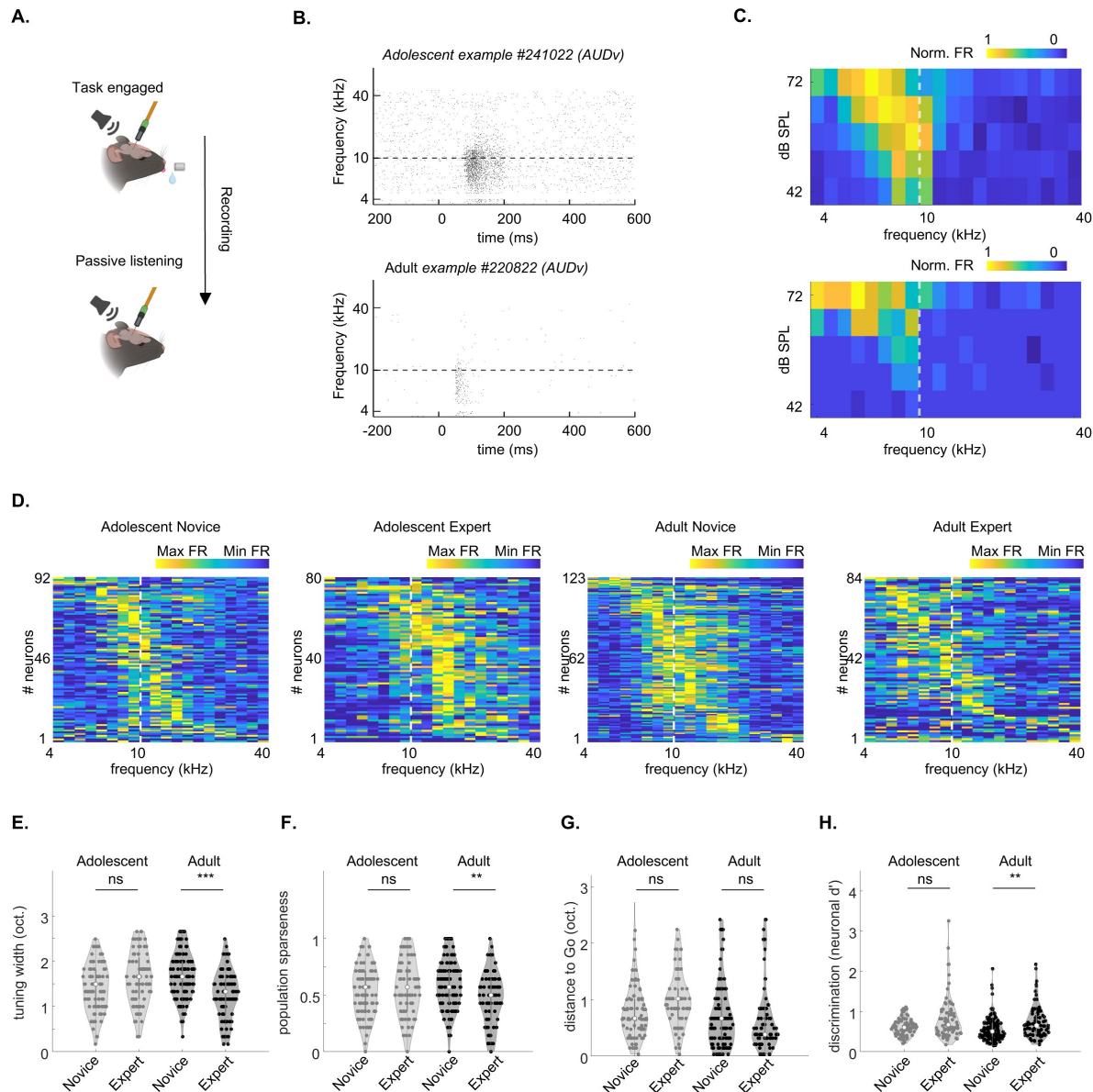


Figure 7

Unlike adults, neuronal tuning properties of adolescents do not change after learning

A. Schematic showing that for the passive listening protocol, we continued our recording following the session of the engaged task (i.e. in satiated mice) by removing the waterspout. Created with *BioRender.com*. **B.** Example raster plot of a neuron from an adolescent mouse (top) and an adult mouse (bottom). **C.** FRA's of the neurons shown in 'B'. **D.** Distribution of best frequencies in our dataset. Values are normalized firing rates calculated at 62 dB SPL. Matrices are sorted by BF for clarity. Dotted line marks the decision boundary. **E.** Tuning bandwidth at 62 dB SPL of neurons in adolescents and adults. Side by side comparisons of novice versus experts. (adolescents $p = 0.0882$, adults $p = 0.0001$, Kruskal Willis Test after Tukey-Kramer correction for multiple comparisons). **F.** Same as E. for Population sparseness (adolescents $p = 0.9549$, adults $p = 0.0013$, Kruskal Willis Test after Tukey-Kramer correction for multiple comparisons). **G.** Same as E. for the distance (in octaves) between the best-frequency of each neuron to the easy Go-stimulus (adolescents $p = 0.0816$, adults $p = 0.6391$, Kruskal Willis Test after Tukey-Kramer correction for multiple comparisons). **H.** Same as E. for the average neuronal d' of frequencies in the learned frequency spectrum (adolescents $p = 0.1627$, adults $p = 0.0026$, Kruskal Willis Test after Tukey-Kramer correction for multiple comparisons).

regions (TEa and AUDd were excluded since less than 20 units responded to the FRA protocol after task engagement, see Supplementary Table 5). We found that the firing properties were distinct between adolescent and adult neurons, in both AUDp and AUDv. However there were no significant differences between AUDp and AUDv in either group (Supplementary Table 6). The peak of frequency tuning was heterogeneous across the frequency range with no clear trends of learning or age (**Fig. 7d**). To quantify these responses, we calculated tuning width (**Fig. 7e**), population sparseness (**Fig. 7f**), the distance between the best frequency of the neurons and the Go frequency of the easy task (**Fig. 7g**), and single neuron discriminability (**Fig. 7h**). As expected from previous work, learning significantly changed neuronal tuning properties in adults (**Fig. 7e-h**, ‘Adults’). Surprisingly, however, learning had no significant effects on any of the single neuron parameters that we tested in the adolescent group (**Fig. 7e-h**, ‘Adolescents’). In adults, these results remained consistent when AUDp and AUDv were analyzed separately (Fig. S7-1). In contrast, adolescent results were more heterogeneous: AUDp exhibited changes in tuning alongside higher discriminability, while AUDv showed no detectable differences. Thus, despite the notion that adolescents’ brains may be more plastic than adults, we found no clear evidence for learning-induced plasticity in basic tuning properties of the adolescent mice following our protocol (Table 5, Fig. S6-1). These results underscore that learning-induced plasticity manifests differently in adolescents and adults.

Discussion

The term “stormy adolescence” aptly captures what we know from psychology, neuroscience, and biology—that adolescence is a period of turbulent development, which shapes the individual’s future behavior and health[33, 34]. While numerous behavioral phenomena of adolescence have been studied, particularly in humans, the neural underpinnings of these remain largely unknown[35]. In this study, we focused on (auditory) cortical neurons of the adolescent brain of mice while they were engaged in an auditory discrimination task. We found numerous differences in behavior, single neuron activity, and population encoding between adolescents and adults.

Adolescence behavior – perceptual and cognitive differences

We utilized the Educage platform to compare between adolescents and adults[18]. The Educage provides an optimal environment for training young mice; its automated nature minimizes human handling, which increases efficiency and reduces to a minimum any interference during the training process. Using the Educage, we found that procedural learning was intact in pre-adolescent mice (P20-P25), and behavioral differences were found only after the mice learned the task during the discrimination phase (P30-37). Compared to adults, adolescent mice scored lower on the hard versions of the task (**Fig. 1i-j**). Interestingly, our data suggests that the lower performance in the hard task in adolescents was not due to a deficiency in perceptual sensitivity (**Fig. 2b**), but rather to a cognitive deficit – namely, response bias (**Fig. 2c**). This result is perhaps not surprising since higher response bias is a cognitive phenomenon that has been well documented in adolescence as a critical period of inhibitory control[36]. Indeed, additional cognitive control mechanisms are still developing during this age [37–39].

Another manifestation of weaker cognitive control mechanisms is the noisy behavioral performance of adolescents (**Fig. 2h-j**). While this noise may reflect higher plasticity and potentially serve cognitive flexibility[6, 40], it poses a disadvantage for the precision and consistency required in our task. This noise may also arise from changes in pubertal hormones, which we did not manipulate[41]. Notably, few studies have explicitly characterized auditory discrimination of adolescents in rodents; the majority of these have been carried out in gerbils using an amplitude modulation (AM) detection task [4, 26, 42]. Despite the different task, many of our age-related findings are aligned with the findings in gerbils, including lower performance in adolescents, and greater variance in adolescent behavior[4]. However, some

findings differ. For example, the response bias which was different in our data, was similar between young and adult gerbils in an AM modulation detection task. This difference may arise from differences in task design, as well as the severity of the punishment used in the task (here, we used a milder punishment). Together, our data support a view that cognitive factors, and to a lesser extent perception of pure tones *per se*, are the primary contributors to the poorer performance observed in adolescent mice when performing a sound discrimination task.

Representation of stimulus and choice in ACx still undergoes maturation in adolescence

Motivated by the abovementioned behavioral differences, we tested how single neurons in ACx—whose involvement in task performance we confirmed via optogenetic silencing in adult mice (Fig. S3-2)—represented sounds (by comparing Hits vs FA) and choices (by comparing FA to CR). While some studies suggest that ACx is not required for simple auditory-guided behaviors [43], others highlight its role in pure tone perception and task execution [19, 20, 44, 45]. While cortical representations of sounds in ACx could be the basis of perceptual constraints, distinct processing of choices would reflect that more cognitive mechanisms are involved. We found that cortical representations of sounds and choices in mice engaged in the task were still immature as compared to adults, e.g., responses in adolescents were more sluggish and less informative (Figs. 4–6). These differences can be a result of age-related differences, and/or the result of a combined ‘age x learning’ effect.

We found modest age-related differences in spiking activity of neurons from layer 5/6 between adolescents and adults (Fig. 6d–g; Fig. 7). This limited difference is perhaps not surprising because the CP for tonal organization in ACx is already complete by the age at which we tested the animals (i.e., on P38 [9, 13]). In contrast, more pronounced differences emerged from the interaction of ‘age x learning’, specifically in mice that learned the task and were actively engaged in the behavior (Fig. 4e, f; Fig. 6d–g). This interaction is evident in the reduced single-neuron discriminability (AUC) during the hard task for expert adolescent mice (Fig. 6d vs. 6f). While this may seem counterintuitive, it is consistent with the idea that learning can drive complex reorganization in sensory representations. For example, we recently showed that learning leads to distinct and sometimes opposing effects on single-neuron responses with respect to whether the task is perceptually easy or difficult [46].

It is important to note that our recordings were restricted to infragranular layers (5/6) and did not sample supragranular layers (e.g., layers 2/3), which may exhibit distinct plasticity profiles [21]. As such, the developmental differences we report may reflect layer-specific properties of deep output layers, particularly their involvement in learning-related plasticity and communication with subcortical or frontal targets. We speculate that immature feedback (FB) to ACx may underlie these differences for several reasons. First, FB from higher cortical regions to more primary cortices has been proposed as a neural substrate for difficult perceptual discriminations [47, 48]. Second, FB activity (often referred to as ‘top down’) is thought to be the key neural pathway involved in cognitive control mechanisms [49]. FB is typically expressed in the late (>100ms) phase of single neurons activity, which was significantly different between adults and adolescents in our data (Fig. 4d). Furthermore, FB may act to stabilize or suppress sensory responses in ACx, reducing apparent single-neuron discriminability of hard stimuli after learning (Fig. 6f), while preserving or enhancing in the population discriminability such as we observed in Fig. 5.

The precise source of FB that contributes to perceptual or cognitive differences remains to be determined, but likely candidates include regions that project to ACx [50, 51], such as medial prefrontal cortex [52, 53], orbitofrontal cortex [54], or anterior cingulate [55]. Testing the causal role of these regions during adolescence will be an important direction for future research.

Cortical coding is performed by populations of neurons, which can efficiently encode stimuli on a trial-by-trial basis[56–60]. Population coding, as measured here by decoding accuracy, largely recapitulated the results of the single neuron data. Specifically, cortical populations in adults encoded the different tones better and faster; particularly in the hard task (Fig. 5a–c). Interestingly, by testing the quality (i.e., separation) of the decoding performance, we revealed a correlation between decoding in the easy and hard tasks in adult (Fig. 5d). This result suggests that the cortical network uses shared representation for decoding the easy and hard tasks, possibly by the same neurons. This is consistent with our recent work showing that indeed the same neurons in ACx respond to the same stimulus differently, depending on if the mouse is engaged in discriminating an easy sound pair or a hard sound pair[46]. The shallower slopes in adolescence suggest a less efficient decoding scheme that may arise from different populations of neurons encoding the stimuli in the easy and hard versions of the task.

Our optogenetic silencing of ACx in expert adult mice resulted in a marked suppression of licking behavior (Fig. S3-2), consistent with the idea that ACx contributes to task performance. However, the strength of the effect raises the possibility that the manipulation may have impacted broader aspects of behavior—such as arousal, motivation, or motor readiness—rather than selectively disrupting auditory discrimination. Recent work [61] suggests that ACx plays a critical role during learning by supporting higher-order computations but may become less essential for task execution once mice reach expert performance. These findings, together with previous reports [19, 20], indicate that the necessity of ACx for auditory-guided behavior may depend on task structure, training stage, and the exact nature of the manipulation. The robust decrease in lick rate and performance observed in our experiment may reflect the intensity, specificity, and timing of the silencing. The use of GtACR2, a highly potent silencer, may have further amplified the effect. We cannot rule out the possibility that silencing the output from ACx also disrupted the flow of information to downstream areas, thereby exerting a stronger influence on perception and behavior. Although the injection was targeted to the ACx, off-target effects are also likely to have contributed. Future experiments using graded or temporally specific perturbations will be useful to more precisely defining the contribution of ACx across learning and development.

Learning in the adolescent brain

Cortical representations of pure tones have a critical period (CP) between P12–P15[13, 16, 17]. Indeed, we chose to study animals at P38–P40 on a pure tone discrimination task precisely to avoid the hypersensitivity associated with this CP. But this may be unavoidable, as recent work has shown that ACx also has a later CP for more complex sounds features[16, 17]. Those late CP's overlap with our training schedule and the underlying biological manifestation of this CP is expected to affect learning related plasticity in general. The underlying mechanism of the late CP to complex sounds has been associated with dynamic changes of inhibitory circuits[62, 63]. Thus, immature inhibitory circuits remain sensitive in adolescents, and could affect how pure tones are represented, despite the closure of the classic tonal CP. Notably, sound discrimination learning changes tonal representations even in adult mice, well after the closure of CP's for both simple and complex sounds [18, 24, 46]. We hypothesize, therefore, that the differences in performance on more difficult discriminations between adolescents and adults are shaped primarily by age-specific learning-related plasticity, rather than by differences in pure tone representation per se.

Taken together, our findings show that cortical responses in adolescents differ from those in adults, despite both groups learning the same pure tone discrimination task under identical training protocols. Although the CPs for pure tones are considered closed by this age, learning to discriminate between tones still varies across developmental stages. It remains unclear how later CPs—such as those shaping responses to amplitude modulation (AM) or frequency-modulated (FM) sweeps through inhibitory circuits—contribute to these differences. Likewise, the role of ongoing maturation in cognitive control, including the development of feedback pathways, warrants further investigation. Nevertheless, it is evident that developmental processes in the

adolescent brain continue to influence how even simple sounds like pure tones are encoded. The adolescent-specific plasticity observed here directly reflects how perceptual and cognitive features are integrated within the auditory cortex (ACx).

Methods

Animals

A total of 47 (37 female and 10 male) C57BL/6 mice were used in this study. Adolescent mice were weaned at postnatal (P) day 20. Adult mice were trained starting from P60. All mice were housed on a 12-hour light/12-hour dark cycle with food and water *ad libitum*, unless used for behavioral training (see below). All experiments were approved by the Institutional Animal Care and Use Committee (IACUC) at the Hebrew University of Jerusalem, Israel (Permit Number: NS-21-16448-4, NS-21-16694-4, NS-22-16966-4).

Surgical procedures

The details for surgical procedures were identical to those used in previous studies from our laboratory [18, 22, 50, 64–67]. The relevant procedures are briefly described below.

RFID Implantation

24h before the start of behavioral training in the automated home-cage, mice were implanted with a radio frequency identification (RFID) chip (Trovan, EXTM). Animals were anesthetized with 2% Isoflurane with pure O₂ as carrier. RFID chips were implanted under the scruff. After implantation, mice were injected with 0.04mg/g of 10% Meloxicam solution to prevent infection and alleviate any possible pain from the implantation.

Head-bar Implantation & Craniotomy

48h before the start behavioral training on the head-fixed recording setup, mice were anesthetized with 2% Isoflurane, the body temperature maintained at 37 °C and the eyes were covered with 5% Chloramphenicol ointment to prevent from drying. Before removal of the scalp, we applied 4% Lidocaine above the skin and skull area. Afterwards, we glued a custom-made titanium bar at the back of the mouse skull using dental cement (Meta-bond). To enable acute recordings in ACx, we performed a craniotomy on the left-hemisphere at 2.5 mm posterior and 4.0 lateral to the bregma. The craniotomy was protected by a pool of dental cement and covered with a silicon elastomer (WPI; Kwik-Cast catalog #KWIK CAST). After the craniotomy, mice were injected with 0.04mg/g of 10% Meloxicam solution and 0.2 ml Saline. During the 48 hours of recovery, mice received two further doses of 0.04mg/g of 10% Meloxicam solution. 24h before recordings, mice were again anesthetized to remove regrowth of the dura-mater during training. The area around the craniotomy was cleaned with Hydrogen-peroxide (3%), the silicon elastomer was replaced, and another dose of 10% Meloxicam was injected.

Virus Injection & Cannula Implantation

The surgical procedure was similar to the *head-bar implantation & craniotomy*. The differences in the procedure are outlined below. Bilateral craniotomies were performed at 2.5 mm posterior and 4.3 mm lateral to the bregma. We injected 200nl AAV5-CAMKII-GtACR2-FRED_kv_2.1, or 200nl AAV9-CAMKII-dTomato. Both viruses were produced at the ELSC virus-core facility. We injected the virus at a depth of 1.1 mm at 0°. Afterwards, we inserted a 0.2 mm diameter optical fiber with an attached cannula (CFML1202; Thorlabs) at a depth of 0.9 mm at 0°, bilaterally. We chronically fixed the fiber position on the skull using dental cement (Meta bond).

Auditory Stimuli

Sound stimuli were delivered from a calibrated free-field speaker (ES1 SN:4568) using a multifunction processor (ED1; Tucker-Davis Technologies). The speakers were calibrated with a free-field microphone (Type 4939, Bruel and Kjaer). The stimuli were comprised of pure tones, sampled at 500 kHz, between 7.07 to 14 kHz for behavioral training and 4-40 kHz for passive listening. All stimuli were played at sound pressure levels of 72 dB SPL for behavioral training, and 72-, 62-, 52-, and 42-dB SPLs for passive listening. All tones were played in randomized order.

Behavior

Automated home-cage training (the ‘EduCage’)

For a detailed description of the EduCage behavioral platform, see [18]. After RFID-implantation, groups of 5 mice were placed in their home-cage that was connected to a behavioral chamber called the EduCage (Fig. 1a). Mice were provided with food *ad libitum*. Water access was restricted to the EduCage chamber. Mice could access the EduCage freely to retrieve water mixed with 5% sucrose. The behavioral training was controlled by a custom LabView 2019 (National Instruments) program running on BNC-2110 (National Instruments) and FPGA (MyRIO – National Instruments) to register licks, RFIDs numbers, and deliver stimuli and reinforcements. Every time a mouse entered the water port its RFID was automatically recognized, a trial initiated, and an auditory stimulus played from a speaker vertically positioned above the behavioral chamber. We trained mice gradually for three weeks on a Go/No-Go paradigm of pure tones, spaced around a category boundary of 10 kHz. This paradigm enabled us to efficiently compare between adolescent and adult behavior, since learning was restricted from post-weaning (P20) to adolescence (P37) (correspondingly adults were trained from P60-P77). Mice were trained in four separate training phases (Fig. 1b). For the first 24h (P20-P21 in the adolescent group of mice and P60-P61 in adult mice) mice were habituated to the behavioral chamber and could approach the port freely to retrieve water after licking the waterspout (0.0015 ml per trial). During the following 48h (P21-P23 in the adolescent group of mice and P61-P63 in adult mice) mice underwent a tone-association phase. All tones played were 300ms long (sampled at 500 kHz). Tone association was restricted to the lowest Go tone (7.07 kHz at 100% probability). Mice received a water-reward in response to licking during the first 2 seconds after tone-offset. This was followed by the first tone-discrimination stage, which contained a No-Go tone (14.14 kHz at 45% probability) at 1 octave away from the Go tone. We analyzed four response outcomes—Hits, Misses, Correct Rejections, False Alarms—in the 2 seconds response window after the tone-offset. ‘Hits’ were counted after 5 successful licks to the Go-tone and followed by a water-reward. ‘Misses’ were counted when licks to the Go-tone trials did not pass the lick threshold. Correspondingly, ‘Correct Reject’ (CR) trials, were all No-Go trials below the lick threshold. CR trials were not reinforced. ‘False Alarms’ (FA) were counted when 5 or more licks were registered after the No-Go. FA trials were followed by a white-noise punishment (5-20 kHz at 72 dB SPLs, 2 seconds) and an inter trial delay of 5 seconds. In addition, mice were exposed to 5 probe trials (tones whose trial outcome that were neither rewarded nor punished at 8.49 kHz, 9.567 kHz, 10 kHz, 10.44 kHz, and 11.89 kHz; played at 2% per tone). Mice trained on the 1 octave tone discrimination for 1 week (P23-P30 in adolescent mice and P63-P70 in adult mice). Next, we added a second pair of tone - a Go/No-Go tone pair spaced 0.25 octaves apart (9.17 kHz and 10.95 kHz) for another week of training. All tones were played in randomized order and trial difficulties of the easy and the hard task were intermingled. The experiment was terminated at age P37 for adolescent and P77 for adult mice.

Head-fixed training

We used a head-fixed paradigm that enabled us to efficiently train adolescent and adult mice on the same task and record from engaged mice after learning. We trained mice on the same Go/No-Go stimulus setup of the Educage described above with few differences which are outlined below.

The head-fixed behavioral training was controlled by a custom MATLAB 2023B (Mathworks) program running on BNC-2110 (National Instruments) to register licks and deliver stimuli and reinforcements. Mice were water-restricted for 24h prior to training, and their weight monitored daily. While training on the setup, mice received water supplemented with 5% sucrose, depending on their weight (0.125 ml per g of body weight). Additional water, in the form of Hydrogel (ClearH2O) was given after daily training if mice did not consume the required amount during training, or their body weight dropped below 85% of the pre-restriction weight. Mice were gradually habituated to the setup by head-fixation and received free water after licking, during the first training day (P23 in adolescent mice and P63 in adult mice). Licking was measured with an optical lick-meter (Sanworks). During the second training day mice were associated with a tone. All tones of the task were 100ms long (sampled at 500 kHz). Mice received a water reward in response to licking in the first 2 seconds after tone offset. Reward-reinforcement was delayed to 0.5 seconds after the tone offset. To break the regularity of the trial sequences, we applied a dynamic inter-trial interval of 6-8 seconds. New trials were initiated if mice did not lick the spout at least 2 seconds prior to the upcoming trial. Mice were familiarized with the tone for at least two training days (P24-26 in adolescent mice and P64-66 in adult mice). Then, and when the lick rate was above 80%, mice proceeded to the first tone discrimination stage. We gradually increased the probability of No-Go trials, depending on the lick rate of the mouse. We added the 0.25 octave Go/No-Go tone pair, after a maximal of seven training days on the 1 octave tone pair (P26-33 in adolescent mice and P66-73 in adult mice), or if the behavioral performance exceeded a threshold of $d' = 1$, as calculated using the signal detection metric. Similar to the Educage, all tones were played in randomized order and trial difficulties of the easy and the hard task were intermingled. Mice were trained until P37 in the adolescent group and P77 in the adult group.

Optogenetic Manipulation

Adult mice (P60) were injected with AAV5-CAMKII-GtACR2-FRED_kv_2.1 ($n = 3$) and implanted with cannulas prior to training as described above. Mice underwent optogenetic stimulation after training on the head-fixed task by attaching a ferrule patch cable (M79L01; Thorlabs) to the implanted cannula using a mating sleeve (ADAF1; Thorlabs). The ferrule cable end (SMA connector) was connected to an LED driver setup (LEDD1B; Thorlabs) that enabled us to send precise outputs at 476 nm (blue light; 5mW) bilaterally. To test the effect of ACx inhibition on task performance, we timed the LED output to 50ms before tone onset until 50ms after tone-offset. The LED consisted of pulses given at 10 Hz for 200ms

Behavioral sessions of optogenetic manipulation were identical to the Go/No-Go paradigm of previous training sessions. We inhibited ACx in 50% of the trials in randomized order. Optogenetic manipulations were repeated throughout multiple sessions of expert performance. As a control, we repeated the same experiment in mice injected with AAV9-CAMKII-dTomato ($n = 3$). After completion of the experiment, animals were deeply anesthetized with Ketamine and medetomidine (0.008 g/kg, and 0.65 mg/kg, respectively) and then perfused with 4% Paraformaldehyde. After the perfusion, the brain was extracted and preserved. We then sectioned the ACx using a cryostat (Leica) into 0.05 mm thick coronal slices. Afterwards brain slices were washed one time under 1% Phosphate-buffered saline (PBS), and a second time with 1% PBS plus 0.4% Triton. Finally, slides were mounted and stained with DAPI (4%) and imaged using a wide-field fluorescent microscope (Olympus Life Science, Olympus IX83), to reconstruct the probe position offline.

Extracellular Recordings

Recording setup

Before the onset of the recording session, we removed the silicon elastomer and placed an external reference electrode (Ag/AgCl wire) on the skull of the right hemisphere. All recordings were performed using Neuropixels 1.0 (Npx; IMEC, phase 3A), together with a base-station connected to a chassis (IMEC; NI PXIe -1071, National Instruments). Probes were mounted to a custom-made steel rod and connected to the ground. Before penetration, probes were covered with a fluorescent dye (Dil (Invitrogen catalog #V22885-red, or Dio (Invitrogen catalog #V22886 – yellow), to reconstruct penetration sites after the recording.

Then, Npx probes were inserted into the left ACx at 30° to a depth of 3.85 mm (this approach limited our recording to deep layers of the ACx). The skull surface was submerged in saline and the probe allowed to settle for 10 min. We sampled recordings at 20 kHz, with action potential band filtered to contain 0.3-to 10kHz frequencies. Action potential band gain was set to 500. Out of the 960 available sites on the 1 cm shank of the Npx probe [68], we acquired the 384 lowest recording shanks in a staggered configuration. We used common-average referencing to process the voltage-traces.

Recording Schedule

We performed recordings of both novice and expert adolescent and adult mice. Novice recordings were performed after tone-association (see *head-fixed training*; adolescent n=3; adult n = 3). Expert recordings were performed after learning (adolescent n= 5; adult n = 6). Both novice and expert recording include the easy (1 octave) and hard (0.25 octave) tone pair. Both recordings were performed after P37 for adolescent mice and P77 for adult mice. Tone probabilities and protocol parameters were identical to the *head-fixed training*. Each mouse was recorded multiple times (novice recordings adolescents n = 6; novice recording adults n =6; expert recordings in adolescents n = 13; expert recordings in adults n = 14). In some mice, after the recording in the engaged configuration, we recorded neural activity under passive listening conditions to a pure tone protocol (novice recordings adolescents n = 5; novice recording adults n = 5; expert recordings in adolescents n = 4; expert recordings in adults n = 4). The pure tone protocol was comprised of 20 different frequencies, logarithmically spaced between 4 and 40 kHz, presented at 5 sound pressure levels (see *Auditory stimuli*). Each frequency and attenuation combination were presented 16 times. The tone interval was set to 1 second. After completion of the recording schedule, animals were deeply anesthetized with Ketamine and medetomidine (0.008 g/kg, and 0.65 mg/kg, respectively) and perfused with 4% Paraformaldehyde. After the perfusion, we performed histological analysis and sectioned coronal brain slices as described above (see *Optogenetic Manipulation*) to reconstruct the exact probe position.

Preprocessing & Spike-sorting

All recordings were sorted using Kilosort 2.5/3 open-source software ([69]; <https://github.com/MouseLand/Kilosort>). After automatic sorting, we performed manual curation of the acquired units using ‘Phy-2’ open-source GUI. (UCL; <https://github.com/cortex-lab/phy>). During manual curation we distinguished between single units (SUs) and noise (pre-labeled multi-units in Kilosort2.5/3 were automatically labelled as noise). The following parameters were set to determine SUs: physiologically plausible waveform shape (1–2 ms, biphasic/triphasic), high spike-amplitude (> 50 μ V, SNR > 3), physiologically plausible refractory period (ISI > 4ms), sufficient inter-spike-interval (ISI peak 5–10 ms), sufficient firing-rate across the recording (>0.1 spikes/sec), and principal-component cluster density (high density and low overlap). In addition, SUs were compared to neighboring units, based on waveform, firing rate, drift-pattern and cross-

correlograms to determine merging of two SUs into a single cluster (Pearson Correlation Coefficient >0.8). Units that passed the above-mentioned criteria were considered single units (SUs).

Data Analysis

The analysis was performed with MATLAB R2023b (MathWorks) and Python3.13 (PyCharm 2024.1.3). In all figures, unless mentioned otherwise, we represent mean and median of the data, together with standard deviation, quartile, or standard error of the mean (MAD/sqrt(n); n = number of data points).

Behavioral analysis

We categorized behavioral responses of the task into hit, miss, false alarm, and correct reject responses. To compare behavioral performance, we calculated the d' value based on the signal detection metric ($d' = \text{the inverse normal distribution of the z-scored hit rate} - \text{the z-scored (false alarm rate)}$). Z-scored values of 0 were rounded to 0.01 and z-scored values of 1 were rounded to 0.99, to avoid d' approaching infinity. Next, we calculated psychometric curves based on the lick rate (above the lick threshold) to learned and probe trials (**Fig. 2b, c, e**). Psychometric curves were normalized and fitted to a sigmoidal function, defined as follows:

$$S(t) = a \frac{1}{1 + e^{\frac{-(t-b)}{c}}} \quad (\text{Equation 1})$$

Where a denotes the lick rate, b = the time of the inflection point, and c = the steepness of the curve. Based on the normalized psychometric curve, we extracted two inflection points (random lick rate at 0.5 and category boundary at 10 kHz). To compare the heterogeneity of adolescent and adult behavior, we calculated the criterion bias (the inverse normal distribution of the $(-0.5 * (\text{z-scored (hit rate)} + \text{z-scored (false alarm rate)}))$) and analyzed the coefficient of variation (CV) of behavior across the learning. We applied similar behavioral measurements to the Educage training and head-fixed training, as well as recordings and optogenetic manipulation during task performance. During expert-performance in the head-fixed task, we analyzed the running d' as the average performance of the last 25 trials (**Fig. 3**). In addition, we also calculated the average lick trace during the reinforcement delay. Here, we also extracted the lick latency per trial per mouse, defined as the lick latency from tone-onset (**Fig. 3**).

Neuronal analysis

Probe trajectories and location of SUs were reconstructed using Allen CCF tools (<https://github.com/cortex-lab/allenCCF>; [70]). For further analysis we only considered SUs that were recorded from infragranular layers 5 and layer 6. SUs from the dorsal-auditory cortex (AUDd), primary auditory cortex (AUDp), ventral auditory cortex (AUDv) and temporal association cortex (TEa) were pooled together as auditory cortex (ACx), and in some analyses presented separately [65]. Neurons were considered excitatory auditory responsive if the average spontaneous firing rate (FR) of the baseline activity preceding the tone (200ms to 50ms before stimulus onset) was significantly lower than the average tone-related activity (from stimulus onset until 50ms after stimulus offset) across all trials [22]. The difference was tested with a right-sided Wilcoxon sign rank test ($p < 0.05$). Suppressed auditory responsive units and non-auditory responsive units were excluded from the analysis. Peri-stimulus time histograms (PSTHs) were smoothed with a gaussian smoothing filter of 5ms. To test the difference between adolescent and adult auditory firing properties in expert mice, we calculated the difference in average spontaneous FR (-200ms preceding the tone onset until tone-onset, average evoked FR (tone onset until 50ms after tone offset) and fraction of responsive trials. We also compared the coefficient of variance of the evoked FR (mean FR divided by standard deviation of the FR across trials). Temporal differences in

firing properties were compared by calculating the latency to peak (latency of the highest FR), minimal latency (first spike after tone-onset), full-width-half-maximum (FWHM; time from peak FR to baseline FR), and the lifetime sparseness, which was calculated as follows:

$$S = \left(1 - \left(\frac{\frac{\sum r_i^2}{n}}{\frac{\sum r_i}{n}} \right) \right) / \left(1 - \frac{1}{n} \right) \quad (\text{Equation 2})$$

r_i = corresponds to the FR of each learned frequency and n equals the number of learned tones. Values of S near 0 correspond to a dense FR and values near 1 to a sparse code [71].

Single neuron analysis during task performance

We investigated the FR of excitatory auditory responses per trial from 200ms preceding the stimulus upto 600ms after the tone onset. To test how well single neurons discriminate between trial outcomes, as well as task difficulty, we calculated task-related activity per trial using receiver operating characteristics (ROC) and calculated the area under the curve (AUC) across a running window of 25ms in bins of 50ms (Fig. 4, Fig. 6). Stimulus-related activity was defined as the hit compared to the false alarm trials, separately for the 1 octave and 0.25 octave tone pair. Stimulus-related activity of miss compared to correct trials were not analyzed due to the lack of miss trials in expert performance. In addition, we also tested the average (mean of 1 octave and 0.25 octave tone pair) choice discrimination, defined as the difference between false alarm and correct reject trials (Fig. 4). We sampled 20 trials per trial outcome and repeated the ROC encoding 100 times. Afterwards, we adjusted the average AUC values of all iterations of each neuron to its baseline AUC before the tone onset ((AUC – baseline AUC) + 0.5). AUC values close to 0.5 indicate a low neuronal discriminability, and values upto 1 indicate high neuronal discriminability [72]. To test significant discrimination, we repeated the ROC encoding with shuffled trial identities (20 randomly shuffled trials per trial outcomes across 100 iterations). Significant discriminability of a neuron was defined as the AUC time bins that exceed the mean ± 3 std of the shuffled distribution [64]. Adolescent and adult neuronal discriminability were compared using the onset-latency of discrimination of a neuron (first timepoint > mean ± 3 std of the shuffled distribution; Fig. 4h, Fig. 6c), the maximal AUC of the running window (Fig. 4j, Fig. 6e) and the duration of discriminability (time duration > mean ± 3 std of the shuffled distribution; Fig. 4i, Fig. 6d). To compare novice and expert performance we repeated the AUC encoding and compared the general discrimination of 1 octave and 0.25 octave Go and No-Go discrimination of all trial outcomes (Fig. 6). To evaluate the correlation of behavioral performance and neuronal discriminability, we compared d' per recording to the average of the maximal AUC of all neurons per recording. The AUC and d' values were compared via pairwise Pearson correlation (Fig. 6f).

Auditory cortex population analysis during task performance

We analyzed the population activity in the ACx of adolescent and adult mice on a trial-by-trial basis. Population activity was segregated by hit and correct-reject trials for both easy and hard tasks in each age group. We excluded recordings that contained fewer than 20 auditory-responsive neurons, fewer than 20 hit or correct-reject trials in either task. This resulted in a total of 8 adolescent sessions and 10 adult sessions. To decode the stimulus from neural activity, we applied linear-discriminant analysis (LDA) to the first 200ms following stimulus onset. Decoding accuracy was calculated on held-out trials (10 trials per stimulus) (Fig. 5a). To examine changes in decoding accuracy over time, we used a 50ms bin width and fitted separate LDA models at each time point, covering a period from -0.5s to 10s after stimulus onset (Fig. 5c). Next, we computed

the separation per session, defined as the ratio of the between-class variance to the within-class variance. Given the means β_1 , β_2 and variances Σ_1 , Σ_2 of the two classes, the separation is given by:

$$S = \frac{\sigma_{between}^2}{\sigma_{within}^2} = \max_w \frac{(w(\mu_1 - \mu_2))^2}{w^T(\Sigma_1 + \Sigma_2)w}. \quad (\text{Equation 3})$$

We used robust linear regression without intercept (Huber loss) to fit the relationship between separations in easy and hard tasks (**Fig. 5d**). To quantify when the decoding accuracy is significantly different from the baseline, we used the bin width with a sample frequency of 100 Hz. Baseline decoding accuracy was calculated as the mean and variance of decoding accuracy during a 500ms window before stimulus onset, up to 50ms before onset. Decoding latency was defined as the time from stimulus onset to the first time point where decoding accuracy exceeded three standard deviations from the baseline mean (**Fig. 5b**). One adolescent session, in which behavioral d' is less than 1, has decoding accuracy that does not exceed three standard deviations from the baseline mean within the first 200 ms. Therefore, this session was excluded from the latency analysis. Finally, we assessed the single-trial variance across both age groups and tasks. Data were standardized per neuron before computing variance per stimulus (**Fig. 5e**). To visualize the results, we projected the population representations to the space spanned by the two LDA decoding dimensions for the easy and hard tasks. We averaged the mean and covariance matrix across sessions for each stimulus and age group, representing the distributions with corresponding ellipses (**Fig. 5f**).

Single neuron analysis during passive listening

We extracted the frequency-response areas (FRAs; **Fig. 7b**) based on the pure-tone responses (from tone onset up to 50ms after tone-offset) of auditory responsive neurons. The bandwidth of excited units was computed for significant excitatory responses of adjacent frequencies at 62 dB SPL and subtracted by the number of expected false-positive responses. Afterwards the sum of significant frequency responses per neuron was multiplied by the octave distance between every frequency to receive an octave-based bandwidth measure. We then calculated the population sparseness as the fraction of significant excited responses in adolescent and adult novice and expert mice (**Fig. 7f**). To test the neuronal discriminability to pure tone responses, we calculated the pairwise d' based on the FR per trial in all frequencies at 62 dB SPL [22, 66]. For two given frequencies p and q , d' values were calculated in the following way:

$$\frac{d(\mu_p, \mu_q)}{\mu(\sqrt{\sum_{i=1}^n (\hat{p}_i - \hat{q}_i)^2})} \quad (\text{Equation 4})$$

Where d represents the distance between the mean FR (β) of frequency p and q , as vectors with n entries representing the mean signal (averaged over individual trials) in n -dimensional space for frequency p/q . The average distance (d) is then divided by the mean of the inner Euclidian distance ($\sqrt{\sum_{i=1}^n (\hat{p}_i - \hat{q}_i)^2}$) between each single trial (**Fig. 7h**).

Statistical Analysis

Statistical comparisons were performed in custom-written codes in MATLAB 2023b (MathWorks) and Python 3.12. We assessed if the data was normally distributed using a Kolmogorov-Smirnov test. Normally distributed data was tested using a paired (within group comparison) or independent (between group comparison) t-test and presented with mean $\pm$ SEM. Non-normally distributed data was tested using a Wilcoxon sign rank (within group comparison), or rank-sum (between group comparison) test. Multiple-samples were tested with ANOVA (parametric data) and Kruskal-Wallis tests (non-parametric data). All tests corrected after multiple comparisons (Bonferroni-correction for two-samples and Tukey-Kramer for multiple samples). In addition, we applied linear-mixed-effect models (LME) to account for hierarchical data structures and

variability within groups of co-housed mice (**Table 1** [↗](#)), repeated measurements of mice and recordings (supplementary Table 1, **Table 3** [↗](#), Table 4), as random effects. Fixed effects, and pairwise interaction effects were adjusted with post-hoc tests and multiple comparisons. For all statistical tests significant differences were defined as p-values below 0.05.

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Additional information

Author Contributions

B.P., and A.M. designed the experiments. B.P, A.D., and A.L., performed the experiments, B.P, F.C., S.D., analyzed the data. B.P., and A.M. wrote the paper with comments from F.C. and S.D.

Inclusion and Diversity

We support inclusive, diverse, and equitable conduct of research

Data availability

The data that support the finding of this study are available through Zenodo (<https://zenodo.org/uploads/13933351> [↗](#)). Additional data is available from authors upon reasonable request.

Code availability

The code is available through GitHub (<https://github.com/benne1295/Age-and-Learning-Shapes-Sound-Representations-in-Auditory-Cortex-During-Adolescence> [↗](#)).

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Additional files

Supplementary figures and tables [↗](#)

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Reviewer #1 (Public review):

Summary:

Praegel et al. explore the differences in learning an auditory discrimination task between adolescent and adult mice. Using freely-moving (EduCage) and head-fixed paradigms, they compare behavioral performance and neuronal responses over the course of learning. The mice were initially trained for seven days on an easy pure frequency tone Go/No-go task (frequency difference of one octave), followed by seven days of a harder version (frequency difference of 0.25 octave). While adolescents and adults showed similar performance on the easy task, adults performed significantly better on the harder task. Quantifying the lick bias of both groups, the authors then argue that the difference in performance is not due to a difference in perception, but rather to a difference in cognitive control. The authors then used neuropixel recordings across 4 auditory cortical regions to quantify the neuronal activity related to the behavior. At the single cell level, the data shows earlier stimulus-related discrimination for adults compared to adolescents in both the easy and hard tasks. At the neuronal population level, adults displayed a higher decoding accuracy and lower onset latency in the hard task as compared to adolescents. Such differences were not only due to learning, but also to age as concluded from recordings in novice mice. After learning, neuronal tuning properties had changed in adults but not in adolescent. Overall, the differences between adolescent and adult neuronal data correlates with the behavior results in showing that learning a difficult task is more challenging for younger mice.

Strengths:

- The behavioral task is well designed, with the comparison of easy and difficult tasks allowing for a refined conclusion regarding learning across age. The experiments with optogenetics and novice mice are completing the research question in a convincing way.
- The analysis, including the systematic comparison of task performance across the two age groups, is most interesting, and reveals differences in learning (or learning strategies?) that are compelling.
- Neuronal recording during both behavioral training and passive sound exposure is particularly powerful, and allows interesting conclusions.

Weaknesses:

- The presentation of the paper must be strengthened. Inconsistencies, missing information or confusing descriptions should be fixed.
- The recording electrodes cover regions in the primary and secondary cortices. It is well known that these two regions process sounds quite differently (for example, one has tonotopy, the other not), and separating recordings from both regions is important to conclude anything about sound representations. The authors show that the conclusions are the same across regions for Figure 4, but is it also the case for the subsequent analysis? Comparing to the original manuscript, the authors have now done the analysis for AuDp and AUDv separately, and say that the differences are similar in both regions. The data however

shows that this is not the case (Fig S7). And even if it were the case, how would it be compatible with the published literature?

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Reviewer #2 (Public review):

Summary:

The authors aimed to find out how and how well adult and adolescent mice discriminate tones of different frequencies and whether there are differences in processing at the level of the auditory cortex that might explain differences in behavior between the two groups. Adolescent mice were found to be worse at sound frequency discrimination than adult mice. The performance difference between the groups was most pronounced when the sounds are close in frequency and thus difficult to distinguish and could, at least in part, be attributed to the younger mice' inability to withhold licking in no-go trials. By recording the activity of individual neurons in the auditory cortex when mice performed the task or were passively listening as well as in untrained mice the authors identified differences in the way that the adult and adolescent brains encode sounds and the animals' choice that could potentially contribute to the differences in behavior.

Strengths:

The study combines behavioural testing in freely-moving and head-fixed mice, optogenetic manipulation and high density electrophysiological recordings in behaving mice to address important open questions about age differences in sound-guided behavior and sound representation in the auditory cortex.

Weaknesses:

For some of the analyses that the authors conducted it is unclear what the rationale behind them is and, consequently, what conclusion we can draw from them.

<https://doi.org/10.7554/eLife.106387.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

A) The presentation of the paper must be strengthened. Inconsistencies, mislabelling, duplicated text, typos, and inappropriate colour code should be changed.

We spotted and corrected several inconsistencies and mislabelling issues throughout the text and figures. Thanks!

B) Some claims are not supported by the data. For example, the sentence that says that "adolescent mice showed lower discrimination performance than adults (l.22) should be rewritten, as the data does not show that for the easy task (Figure 1F and Figure 1H).

We carefully reviewed the specific claims and fixed some of the wording so it adheres to the data shown.

C) In Figure 7 for example, are the quantified properties not distinct across primary and secondary areas?

We now carried out additional analysis to test this. We found that while AUDp and AUDv exhibit distinct tuning properties, they show similar differences between adolescent and adult neurons (see Supplementary Table 6, Fig. S7-1a-h). Note that TEa and AUDd could not be evaluated due to low numbers of modulated neurons in this protocol.

D) Some analysis interpretations should be more cautious. (...) A lower lick rate in general could reflect a weaker ability to withhold licking- as indicated on I.164, but also so many other things, like a lower frustration threshold, lower satiation, more energy, etc).

That is a fair comment, and we refined our interpretations. Moreover, we also addressed whether impulsiveness impacted lick rates. In the Educage, we found that adolescent mice had shorter ITIs only after FAs (Fig. S2-1). In the head-fixed setup, we examined (1) the proportion of ITIs where licks occurred (Fig. S3-1c) and (2) the number of licks in these ITIs (Fig. S3-1d). We found no differences between adolescents and adults, indicating that the differences observed in the main task are not due to general differences in impulsiveness (Fig. S2-1, Fig. S3-1c, d). Finally, we note that potential differences in satiation were already addressed in the original manuscript by carefully examining the number of trials completed across the session. See also Review 3, comment #1 below.

Reviewer #2 (Public review):

A) For some of the analyses that the authors conducted it is unclear what the rationale behind them is and, consequently, what conclusion we can draw from them.

We reviewed the manuscript carefully and revised the relevant sections to clarify the rationale behind the analyses. See detailed responses to all the reviewer's specific comments.

B) The results of optogenetic manipulation, while very interesting, warrant a more in-depth discussion.

We expanded our discussion on these experiments (L495-511) and also added an additional analysis to strengthen our findings (Fig. S3-2e).

Reviewer #3 (Public review):

(1) The authors report that "adolescent mice showed lower auditory discrimination performance compared to adults" and that this performance deficit was due to (among other things) "weaker cognitive control". I'm not fully convinced of this interpretation, for a few reasons. First, the adolescents may simply have been thirstier, and therefore more willing to lick indiscriminately. The high false alarm rates in that case would not reflect a "weaker cognitive control" but rather, an elevated homeostatic drive to obtain water. Second, even the adult animals had relatively high (~40%) false alarm rates on the freely moving version of the task, suggesting that their behavior was not particularly well controlled either. One fact that could help shed light on this would be to know how often the animals licked the spout in between trials. Finally, for the head-fixed version of the task, only d' values are reported. Without the corresponding hit and false alarm rates (and frequency of licking in the intertrial interval), it's hard to know what exactly the animals were doing.

First, as requested, we added the Hit rates and FA rates for the head-fixed task (Fig. S3-1a). Second, as requested by the reviewer, we performed additional analyses in both the Educage

and head-fixed versions of the task. Specifically, we analyzed the ITI duration following each trial outcome. We found that adolescent mice had shorter ITIs only after Fas (Fig. S2-1). In the head-fixed setup, we examined (1) the proportion of ITIs during which licks occurred (Fig. S3-1c) and (2) the number of licks in these ITIs (Fig. S3-1d). We found no differences between adolescents and adults, indicating that the differences observed in the main task are not due to general differences in impulsiveness (Fig. S2-1, Fig. S3-1c, d). See also comment #D of reviewer #1 above.

B) There are some instances where the citations provided do not support the preceding claim. For example, in lines 64-66, the authors highlight the fact that the critical period for pure tone processing in the auditory cortex closes relatively early (by ~P15). However, one of the references cited (ref 14) used FM sweeps, not pure tones, and even provided evidence that the critical period for this more complex stimulus occurred later in development (P31-38). Similarly, on lines 72-74, the authors state that "ACx neurons in adolescents exhibit high neuronal variability and lower tone sensitivity as compared to adults." The reference cited here (ref 4) used AM noise with a broadband carrier, not tones.

We carefully checked the text to ensure that each claim is accurately supported by the corresponding reference.

C) Given that the authors report that neuronal firing properties differ across auditory cortical subregions (as many others have previously reported), why did the authors choose to pool neurons indiscriminately across so many different brain regions?

We appreciate the reviewer's concern. While we acknowledge that pooling neurons across auditory cortical subregions may obscure region-specific effects, our primary focus in this study is on developmental differences between adolescents and adults, which were far more pronounced than subregional differences.

To address this potential limitation: (1) We analyzed firing differences across subregions during task engagement (see Fig. S4-1, S4-2, S4-3; Supplementary Tables 2 and 3). (2) We have now added new analyses for the passive listening condition in AUDp and AUDv (Fig. S7-1; Supplementary Table 6).

These analyses support our conclusion that developmental stage has a greater impact on auditory cortical activity than subregional location in the contexts examined. For clarity and cohesion, the main text emphasizes developmental differences, while subregional analyses are presented in the Supplement.

D) And why did they focus on layers 5/6? (Is there some reason to think that age-related differences would be more pronounced in the output layers of the auditory cortex than in other layers?)

We agree that other cortical layers, particularly supragranular layers, are important for auditory processing and plasticity. Our focus on layers 5/6 was driven by both methodological and biological considerations. Methodologically, our electrode penetrations were optimized to span multiple auditory cortical areas, and deeper layers provided greater mechanical stability for chronic recordings. Biologically, layers 5/6 contain the principal output neurons of the auditory cortex and are well-positioned to influence downstream decision-making circuits. We acknowledge the limitation of our recordings to these layers in the manuscript (L268; L464-8).

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) The presentation of the paper must be strengthened. As it is now, it makes it difficult to appreciate the strengths of the results. Here are some points that should be addressed:

a) The manuscript is full of inconsistencies that should be fixed to improve the reader's understanding. For example, the description on l.217 and the Figure. S3-1b, the D' value of 0 rounded to 0.01 on l. 735 (isn't it rather the z-scored value that is rounded? A D' of 0 is not a problem), the definition of lick bias on l. 750 and the values in Fig.2, the legend of Figure 7F and what is displayed on the graph (is it population sparseness or responsiveness?), etc.

We adjusted the legend and description of former Fig. S3-1b (now Fig. S3-2b).

We now clarify that the rounded values refer to z-scored hit and false alarm rates that we used in the d' calculation. We adjusted the definition of the lick bias in Fig. 2 and Fig. S3-1b (L804).

We replaced 'population responsiveness' with 'population sparseness' throughout the figures, legend and the text.

b) References to figures are sometimes wrong (for example on l. 737,739).

c) Some text is duplicated (for example l. 814 and l. 837).

d) Typos should be corrected (for example l. 127, 'the', l. 787, 'upto').

We deleted the incorrect references of this section, removed the duplicated text, and corrected the typos.

e) Color code should be changed (for example the shades of blue for easy and hard tasks - they are extremely difficult to differentiate).

After consideration, we decided to retain the blue color code (i.e., Fig. 1d, Fig. 3d, Fig. 4e-g, Fig. 5c, Fig. 6d-g), where the distinction between the shades of blue appears sufficiently clear and maintains visual consistency and aesthetic appeal. We did however, made changes in the other color codes (Fig. 4, Fig. 5, Fig. 6, Fig. 7).

f) Figure design should be improved. For example, why is a different logic used for displaying Figure 5A or B and Figure 1E?

We adjusted the color scheme in Fig. 5. We chose to represent the data in Fig. 5 according to task difficulty, as this arrangement best illustrates the more pronounced deficits in population decoding in adolescents during the hard task.

g) Why use a 3D representation in Figure 4G? (2)

The 3D representation in Fig. 4g was chosen to illustrate the 3-way interactions between onset-latency, maximal discriminability, and duration of discrimination.

g) Figure 1A, lower right panel- should "response" not be completed by "lick", "no lick"?

We changed the labels to "Lick" and "No Lick" in Fig. 1a.

h) l.18 the age mentioned is misleading, because the learning itself actually started 20 days earlier than what is cited here.

Corrected.

i) Explain what AAV5... is on l.212.

We added an explanation of virus components (see L216-220).

(2) The comparison of CV in Figure 2 H-J is interesting. I am curious to know whether the differences in the easy and hard tasks could be due to a decrease in CV in adults, rather than an increase in CV in adolescents? Also, could the difference in J be due to 3 outliers?

We agree that the observed CV differences may reflect a reduction in variability in adults rather than an increase in adolescents. We have revised the Results section accordingly to acknowledge this interpretation.

Regarding the concern about potential outliers in Fig. 2J, we tested the data for outliers using the `isoutlier` function in MATLAB (defining outliers as values exceeding three standard deviations from the mean) and found no such cases.

(3) Figure 2c shows that there is no difference in perceptual sensitivity between adolescents and adults, whereas the conclusion from Figure 4 is that adolescents exhibit lower discriminability in stimulus-related activity. Aren't these results contradictory?

This is a nuanced point. The similar slopes of the psychometric functions (Fig. 2c) indicating comparable perceptual sensitivity and the lower AUC observed in the ACx of adolescents (Fig. 4) do not necessarily contradict each other. These two measures capture related but distinct issues: psychometric slopes reflect behavioral output, which integrates both sensory encoding and processing downstream to ACx, while the AUC analysis reflects stimulus-related neural activity in ACx, which may still include decision-related components.

Note that stimulus-related neural discriminability outside the context of the task is not different between adolescent and adult experts (Fig. 7h; $p = 0.9374$, Kruskal Willis Test after Tukey-Kramer correction for multiple comparisons; not discussed in the manuscript). This suggests that there are differences that emerge when we measure during behavior. Also note that behavior may rely on processing beyond ACx, and it is possible that downstream areas compensate for weaker cortical discriminability in adolescents — but this issue merits further investigation.

(4) Why do you think that the discrimination in hard tasks decreases with learning (Figure 6D vs Figure 6F)?

This is another nuanced point, and we can only speculate at this stage. While it may appear counterintuitive that single-neuron discriminability (AUC) for the hard task is reduced after learning (Fig. 6D vs. 6F), we believe this may reflect a shift in sensory coding in expert animals. In a recent study (Haimson et al., 2024; Science Advances), we found that learning alters single-neuron responses in the easy versus hard task in complex and distinct ways, which may account for this result. It is also possible that, in expert mice, top-down mechanisms such as feedback from higher-order areas act to suppress or stabilize sensory responses in auditory cortex, reducing the apparent stimulus selectivity of single neurons (e.g., AUC), even as behaviorally relevant information is preserved or enhanced at the population level.

Reviewer #2 (Recommendations for the authors):

This is very interesting work and I enjoyed reading the manuscript. See below for my comments, queries and suggestions, which I hope will help you improve an already very good paper.

We thank the reviewer for the meticulous and thoughtful review.

(1) Line 107: x-axis of panel 1e says 'pre-adolescent'.

(2) Line 130: replace 'less' with 'fewer'.

(3) Line 153: 'both learned and catch trials': I find the terminology here a bit confusing. I would typically understand a catch trial to be a trial without a stimulus but these 'catch' trials here have a stimulus. It's just that they are not rewarded/punished. What about calling them probe trials instead?

We corrected the labelling (1), reworded to 'fewer' and 'probe trials' (2,3).

(4) Line 210: The results of the optogenetics experiments are very interesting. In particular, because the effect is so dramatic and much bigger than what has been reported in the literature previously, I believe. Lick rates are dramatically reduced suggesting that the mice have pretty much stopped engaging in the task and the authors very rightly state that the 'execution' of the behavior is affected. I think it would be worth discussing the implications of these results more thoroughly, perhaps also with respect to some of the lesion work. Useful discussions on the topic can be found, for instance, in Otchy et al., 2015; Hong et al., 2018; O'Sullivan et al., 2019; Ceballo et al., 2019 and Lee et al., 2024. Are the mice unable to hear anything in laser trials and that is why they stopped licking? If they merely had trouble distinguishing them then we would perhaps expect the psychometric curves to approach chance level, i.e. to be flat near the line indicating a lick rate of 0.5. Could the dramatic decrease in lick rate be a motor issue? Can we rule out spillover of the virus to relevant motor areas? (I understand all of the 200nL of the virus were injected at a single location) Or are the effects much more dramatic than what has been reported previously simply because the GtACR2 is much more effective at silencing the auditory cortex? Could the effect be down to off-target effects, e.g. by removing excitation from a target area of the auditory cortex, rather than the disruption of cortical processing?

We have now expanded the discussion in the manuscript to more thoroughly consider alternative interpretations of the strong behavioral effect observed during ACx silencing (L495–511). In particular, we acknowledge that the suppression of licking may reflect not only impaired sensory discrimination but also broader disruptions to arousal, motivation, or motor readiness. We also discuss the potential impact of viral spread, circuit-level off-target effects, and the potency of GtACR2 as possible contributors. We highlight the need for future work using more graded or temporally precise manipulations to resolve these issues.

(5) Line 226: Reference 19 (Talwar and Gerstein 2001) is not particularly relevant as it is mostly concerned with microstimulation-induced A1 plasticity. There are, however, several other papers that should be cited (and potentially discussed) in this context. In particular, O'Sullivan et al., 2019 and Ceballo et al., 2019 as these papers investigate the effects of optogenetic silencing on frequency discrimination in head-fixed mice and find relatively modest impairments. Also relevant may be Kato et al., 2015 and Lee et al., 2024, although they look at sound detection rather than discrimination.

We changed the references and pointed the reader to the (new section) Discussion.

(6) Line 253: 'engaged [in] the task.

(7) Figure 4: It appears that panel S4-1d is not referred to anywhere in the main text.

Fixed.

(8) Line 260: Might be useful to explain a bit more about the motivation behind focusing on L5/L6. Are there mostly theoretical considerations, i.e. would we expect the infragranular layers to be more relevant for understanding the difference in task performance? Or were there also practical considerations, e. g. did the data set contain mostly L5/L6 neurons because those were easier to record from given the angle at which the probe was inserted? If those kinds of practical considerations played a role, then there is nothing wrong with that but it would be helpful to explain them for the benefit of others who might try a similar recording approach.

There were no deep theoretical considerations for targeting L5/6. Our focus on layers 5/6 was driven by both methodological and biological considerations. Methodologically, our electrode penetrations were optimized to span multiple auditory cortical areas, and deeper layers provided greater mechanical stability for chronic recordings. Biologically, layers 5/6 contain the principal output neurons of the auditory cortex and are well-positioned to influence downstream decision-making circuits. We acknowledge the limitation of our recordings to these layers in the manuscript (L268; L463–467). See also comment D of reviewer 3.

(9) Supplementary Table 2: The numbers in brackets indicate fractions rather than percentages.

Fixed.

(10) Figure S4-3: The figure legend implies that the number of neurons with significant discriminability for the hard stimulus and significant discriminability for choice was identical. (adolescent neurons = 368, mice = 5, recordings = 10; adult $n = 544$, mice = 6, recordings = 12 in both cases). Presumably, that is not actually the case and rather the result of a copy/paste operation gone wrong. Furthermore, I think it would be helpful to state the fractions of neurons that can discriminate between the stimuli and between the choices that the animal made in the main text.

Thank you for spotting the mistake. We corrected the n's and added the percentage of neurons that discriminate stimulus and choice in the main text and the figure legend.

(11) Line 301: 'We used a ... decoder to quantify hit versus correct reject trial outcomes': I'm not sure I understand the rationale here. For the single unit analysis hit and false alarm trials were compared to assess their ability to discriminate the stimuli. FA and CR trials were compared to assess whether neurons can encode the choice of the mice. But the hit and CR trials which are contrasted here differ in terms of both stimulus and behavior/choice so what is supposed to be decoded here, what is supposed to be achieved with this analysis?

Thank you for this important point. You're correct that comparing hit and CR trials captures differences in both stimulus and choice, or task-related differences. We chose this contrast for the population decoding analysis to achieve higher trial counts per session and similar number of trials which are necessary for the reliability of the analysis. While this approach does not isolate stimulus from choice encoding, it provides an overall measure of how well

population activity distinguishes task-relevant outcomes. We explicitly acknowledge this issue in L313-314.

(12) Line 332: *What do you mean when you say the novice mice were 'otherwise fully engaged' in the task when they were not trained to do the task and are not doing the task?*

By "otherwise fully engaged," we mean that novice mice were actively participating in the task environment, similar to expert mice — they were motivated by thirst and licked the spout to obtain water. The key distinction is that novice mice had not yet learned the task rules and likely relied on trial-and-error strategies, rather than performing the task proficiently.

(13) Line 334: *'regardless of trial outcome': Why is the trial outcome not taken into account? What is the rationale for this analysis? Furthermore, in novice mice a substantial proportion of the 'go' trials are misses. In expert mice, however, the proportion of 'miss trials' (and presumably false alarms) will by definition be much smaller. Given this, I find it difficult to interpret the results of this section.*

This approach was chosen to reliably decode a sufficient number of trials for each task difficulty (i.e. expert mice predominantly performed CRs on No-Go trials and novice mice often showed FAs). Utilizing all trial outcomes ensured that we had enough trials for each stimulus type to accurately estimate the AUCs. This approach avoids introducing biases due to uneven trial numbers across learning stages.

(14) Line 378: *'differences between adolescents and adults arise primarily from age': Are there differences in any of the metrics shown in 7e-h between adolescents and adults?*

We confirm that differences between adolescents and adults are indeed present in some metrics but not others in Figure 7e–h. Specifically, while tuning bandwidth was similar in novice animals, it was significantly lower in adult experts (Fig. 7e; novice: $p = 0.0882$; expert: $p = 0.0001$ Kruskal Willis Test after Tukey-Kramer correction for multiple comparisons; not discussed in the manuscript). The population sparseness was similar in both novice and expert adolescent and adult neurons (Fig. 7f; novice: $p = 0.2873$; expert: $p = 0.1017$, Kruskal Willis Test after Tukey-Kramer correction for multiple comparisons; not discussed in the manuscript). The distance to the easy go stimulus was similar in novice animals, but lower in adult experts (Fig. 7g; novice: $p = 0.7727$; expert: $p = 0.0001$, Kruskal Willis Test after Tukey-Kramer correction for multiple comparisons; not discussed in the manuscript). The neuronal d-prime was similar in both novice and expert adolescent and adult neurons (Fig. 7h; novice: $p = 0.7727$; expert: $p = 0.0001$, Kruskal Willis Test after Tukey-Kramer correction for multiple comparisons; not discussed in the manuscript).

(15) Line 475: *'...well and beyond...': something seems to be missing in this statement.*

(16) Line 487: *'onto' should be 'into', I think.*

(17) Line 610 and 613: *'3 seconds' ... '2.5 seconds': Was the response window 3s or 2.5s?*

(18) Line 638: *'set' should be 'setup', I believe.*

All the mistakes mentioned above, were fixed. Thanks.

(19) Line 643: *'Reward-reinforcement was delayed to 0.5 seconds after the tone offset': Presumably, if they completed their fifth lick later than 0.5 seconds after the tone, the reward delivery was also delayed?*

Apologies for the lack of clarity. In the head-fixed version, there was no lick threshold. Mice were reinforced after a single lick. If that lick occurred after the 0.5-second reinforcement delay following tone offset, the reward or punishment was delivered immediately upon licking.

(20) Line 661: 'effect [of] ACx'.

(21) Line 680: 'a base-station connected to chassis'. The sentence sounds incomplete.

(22) Line 746: 'infliction', I believe, should say 'inflection'.

(23) Line 769: 'non-auditory responsive units': Shouldn't that simply say 'non-responsive units'? The way it is currently written I understand it to mean that these units were responsive (to some other modality perhaps) but not to auditory stimulation.

(24) Line 791: 'bins [of] 50ms'.

(25) Line 811: 'all of' > 'of all'.

(26) Line 814: Looks like the previous paragraph on single unit analysis was accidentally repeated under the wrong heading.

(27) Line 817: 'encoded' should say 'calculated', I believe.

All the mistakes mentioned above were fixed. Thanks.

(28) Line 869: 'bandwidth of excited units': Not sure I understand how exactly the bandwidth, i.e. tuning width was measured.

We acknowledge that our previous answer was unclear and expanded the Methods section. To calculate bandwidth, we identified significant tone-evoked responses by comparing activity during the tone window to baseline firing rates at 62 dB SPL ($p < 0.05$). For each neuron, we counted the number of contiguous frequencies with significant excitatory responses, subtracting isolated false positives to correct for chance. We then converted this count into an octave-based bandwidth by multiplying the number of frequency bins by the octave spacing between them (0.1661 octaves per step).

(29) Line 871: 'population sparseness': Is that the fraction of tone frequencies that produced a significant response? I would have thought that this measure is very highly correlated to your measure of bandwidth, to the point of being redundant, but I may have misunderstood how one or the other is calculated. Furthermore, the Y label of Figure 7f says 'responsiveness' rather than sparseness and that would seem to be the more appropriate term because, unless I am misunderstanding this, a larger value here implies that the neuron responded to more frequencies, i.e. in a less sparse manner.

We have clarified the use of the term "population sparseness" and updated the Y-axis label in Figure 7f to better reflect this measure. This metric reflects the fraction of tone–attenuation combinations that elicited a significant excitatory response across the entire population of neurons, not within individual units.

While this measure is related to bandwidth, it captures a distinct property of the data. Bandwidth quantifies how broadly or narrowly a single neuron responds across frequencies at a fixed intensity, whereas population sparseness reflects how distributed responsiveness is across the population as a whole. Although the two measures are related, since broadly tuned neurons often contribute to lower population sparseness, they capture distinct aspects of neural coding and are not redundant.

(30) Line 881: I think this line should refer to Figure 7h rather than 7g.

Fixed.

Reviewer #3 (Recommendations for the authors):

(1) In the Educage, water was only available when animals engaged in the task; however, there is no mention of whether/how animal weight was monitored.

In the Educage, mice had continuous access to water by voluntarily engaging in the task, which they could perform at any time. Although body weight was not directly monitored, water access was essentially ad libitum, and mice performed hundreds of trials per day, thereby ensuring sufficient daily intake. This approach allowed us to monitor hydration (ad libitum food is supplied in the home cage). The 24/7 setup, including automated monitoring of trial counts and water consumption, was reviewed and approved by our institutional animal care and use committee (IACUC).

(2) In Figure 2B-C and Figure 2E, the y-axis reads "lick rate". At first glance, I took this to mean "the frequency of licking" (i.e. an animal typically licks at a rate of 5 Hz). However, what the authors actually are plotting here is the proportion of trials on which an animal elicited ≥ 5 licks during the response window (i.e. the proportion of "yes" responses). I recommend editing the y-axis and the text for clarity.

We replaced the y-label and adjusted the figure legend (Fig. 2).

(3) I didn't see any examples of raw (filtered) voltage traces. It would be worth including some to demonstrate the quality of the data.

We have added an example of a filtered voltage trace aligned to tone onset in Fig. S4-1a to illustrate data quality. In addition, all raw and processed voltage traces, along with relevant analysis code, are available through our GitHub repository and the corresponding dataset on Zenodo.

(4) The description of the calculation of bias (C) in the methods section (lines 749-750) is incorrect. The correct formula is $C = -0.5 * [z(\text{hit rate}) + z(\text{fa rate})]$. I believe this is the formula that the authors used, as they report negative C values. Please clarify or correct.

Thanks for spotting this. It is now corrected.

(5) The authors use the terms 'naïve' and 'novice' interchangeably. I suggest sticking with one term to avoid potential confusion.

(6) Multiple instances: "less trials/day" should be "fewer trials/day"

(7) Supplementary Table 2: The values reported are proportions, not percentages. Please correct.

(8) Line 270: Table 2 does not show the number of neurons in the dataset categorized by region. Perhaps the authors meant Supplementary Table 2?

Fixed. Thank you for pointing these mistakes out.

(9) Figure 5C: the data from the hard task are entirely obscured by the data from the easy task. I recommend splitting it into two different plots.

We agree and split the decoding of the easy and the hard task into two graphs (left: easy task; right: hard task). Thank you!

(10) How many mice contributed to each analyzed data set? Could the authors provide a breakdown in a table somewhere of how many neurons were recorded in each mouse and which ones were included in which analyses?

We added an overview of the analyzed datasets in supplementary Table 7. Please note that the number of mice and neurons used in each analysis is also reported in the main text and legends. Importantly, all primary analyses were conducted using LME models, which explicitly account for hierarchical data structure and inter-mouse variability, thereby addressing potential concerns about data imbalance or bias.

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